ATLANTECH

INTELLIGENT PORT OPERATIONS

Master Roadmap

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Master Roadmap

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SECTION 1

How to read this plan

The work is split into 8 sprints. Every task is assigned to a team only; each team lead decides who in the team does it and how.

Rules

- The Backend and UI/UX teams work ahead: they prepare the APIs and screens before the sprint in which they are used.
- From Sprint 2, every sprint opens with a **Feature Readiness Review (FRR)**: the team leads check the APIs and screens that are ready, approve them and hand them to the Frontend and Mobile teams.
- The Frontend and Mobile teams never build on an API or screen that was not approved. The only exception is independent work: project setup, shared components, static pages and loading, empty and error pages.
- From Sprint 2, every sprint closes with QA release testing on the release branch, a sprint review and a tagged release (develop → release → main, tag, PortHub, demo). Sprint 1 has no release.
- Every contract (CT) is approved in Sprint 1. Each open decision (DEC) is closed in the sprint where the first task that needs it starts, at that sprint's FRR or earlier. Details are in the Decisions file and the Contracts file.
- The UI/UX Team checks the screens built in every sprint against the designs and then updates the designs from the review.
- Git: every task has a branch **feature/<task-id>** from develop and returns through a pull request reviewed by the team lead and the project lead. main and release are protected; only the leader merges into them.

Task tables

Column	Meaning
ID	The task number. The same number is used in the tracker and in the branch name.
Task	What must be delivered and when it counts as done. How to do it is up to the team.
Needs from own team	Tasks of the same team that must finish first.
Needs from other teams	What the team must receive from another team, with the delivering team next to each ID.

Every task ID in this document is a link: select it to jump to that task.

Features

ID	Feature
FT-01	Platform foundation and contracts
FT-02	Sign-in and session
FT-03	Access control, users, devices and audit
FT-04	Port master data and safety limits
FT-05	Department and terminal master data
FT-06	Berthing requests and BAPLIE
FT-07	Customs approval and entry window (AI Part 1)

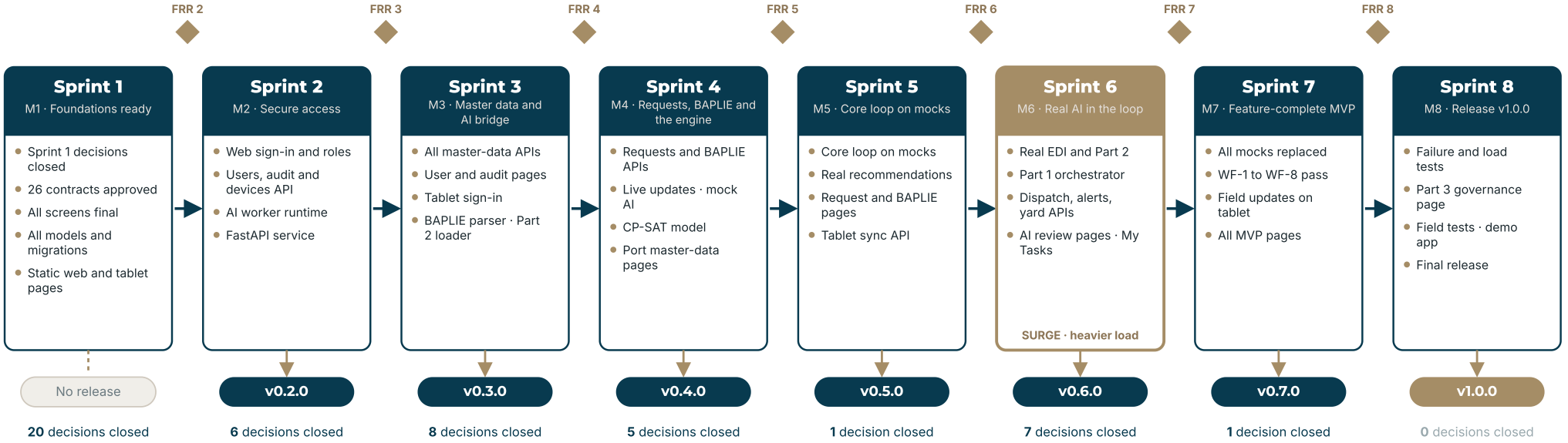
ID	Feature
FT-08	Recommendation review and plan (AI Part 2)
FT-09	Dispatch, notifications and field execution
FT-10	Deviations, re-plan and monitoring
FT-11	Yard placement and TOS hand-off
FT-12	Model evaluation and drift (AI Part 3)
FT-13	Integration, quality and release

SECTION 2

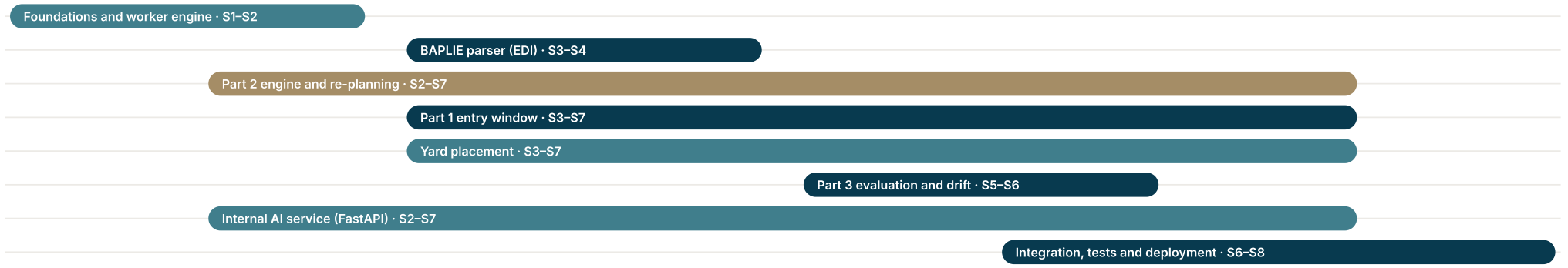
Delivery path

From foundations to v1.0.0

Each sprint opens with a Feature Readiness Review (◆ FRR) from Sprint 2 and closes with a tagged release.



AI track



M1 · Foundations ready

Sprint 1 — Decisions, contracts and foundations

83 tasks
No release

Goal. Close the decisions needed first and approve every contract between the teams. The UI/UX Team finalises every screen, the Backend Team finishes all data models and migrations, and the Frontend and Mobile teams build their foundations and the static pages. There is no release in this sprint.

Handed off at the FRR

Hand-off	Approved items	To
Independent work only	No approved API exists yet; Frontend and Mobile build foundations and static pages.	Frontend · Mobile

Done when

- Every decision due in Sprint 1 is closed and recorded
- Contracts CT-01 to CT-26 are approved in PortHub
- All final screens (UX-W1 to UX-W6) are accepted
- All models and migrations are merged and every pipeline is green
- Static web and tablet pages are done

Leads Council · 1 tasks

ID	Task	Needs from own team	Needs from other teams
PL-01	**Kick-off: present and adopt the Master Roadmap** Present the Master Roadmap to all teams, confirm the way of working and the Definition of Done, and open Sprint 1. Done when every team lead has accepted the plan.	—	—

AI Team · 25 tasks

ID	Task	Needs from own team	Needs from other teams
AI-010	**Module structure in project-ai (common, edi, part1, part2, part3)** Organise the AI repository into one shared part and one part per worker (BAPLIE parser, Part 1, Part 2, Part 3) so each worker can grow on its own. Done when every part has its place and the shared part is reused.	—	—
AI-014	**Unit, datum and time utilities** Build shared helpers for units, depth reference levels and time zones so every part converts values the same way. Done when all conversions come from one place and are tested.	AI-010	—
AI-016	**CI for AI code (lint, types, tests with MySQL and Redis, secret scan)** Check every AI push automatically: style, types, tests against a real database and Redis, and no secrets in the code. Done when a failing check blocks a pull request.	AI-010	—
DEC-03	**Close DEC-03: Location of the shared validation rule sets (override, yard adjustment, BAPLIE correction, window re-check)** The same hard rules must apply in two places: when the AI generates a result and when a user edits it. The teams disagree: the AI wants a background job, the Backend wants the check inside Laravel. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	DEC-10	—

AI Team · 25 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-04	**Close DEC-04: Job envelope, versioning, retries and stale-run timeout** The default Laravel queue format serialises PHP classes that Python cannot read. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-06	**Close DEC-06: Joint solve scope: which requests may change per trigger** CP-SAT solves for the whole port, so we must define what is fixed and what is variable. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-10	**Close DEC-10: Scope of the AI FastAPI service** The AI will expose APIs with FastAPI. The documents say Laravel and Python talk only through Redis, so the role of FastAPI must be set. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-12	**Close DEC-12: AI port_setting keys** AI settings must change without code changes. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-14	**Close DEC-14: C-01 · Part 1 run** The message that starts the entry-window computation for one request on queue:part1. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	DEC-04	—
DEC-15	**Close DEC-15: C-02 · Part 2 solve** The message for solve, regenerate, resource change and delay re-plan on queue:part2. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	DEC-04 DEC-06	—
DEC-16	**Close DEC-16: C-03 · Yard placement** The yard recommendation message on queue:part2. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	DEC-04	—
DEC-17	**Close DEC-17: C-04 · BAPLIE parse** The BAPLIE parsing message on queue:edi. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	DEC-04	—

AI Team · 25 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-18	**Close DEC-18: C-05 · Validation** The shape of the validation request and its result. Depends on DEC-03 and DEC-10. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	DEC-03 DEC-10	—
DEC-19	**Close DEC-19: C-06 · Completion message** The shape of the completion message the worker sends to the Backend. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	DEC-01 (Backend)
DEC-23	**Close DEC-23: Objective terms, weights and number of options** CP-SAT needs a clear objective. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
CT-08	**Contract: AI job envelope (JSON Schema)** Write the contract for the common message format for all work sent to the AI workers, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	DEC-04	—
CT-09	**Contract: C-01 Part 1 run** Write the contract for the message that starts an entry-window calculation and the results it writes, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	CT-08 DEC-14	—
CT-10	**Contract: C-02 Part 2 solve** Write the contract for the message that starts a berth recommendation, a regeneration, a resource-change refresh or a delay re-plan, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	CT-08 DEC-15	—
CT-11	**Contract: C-03 Yard placement** Write the contract for the message that starts a yard placement and the results it writes, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	CT-08 DEC-16	—
CT-12	**Contract: C-04 BAPLIE parse** Write the contract for the message that starts BAPLIE parsing and the results it writes, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	CT-08 DEC-17	—
CT-13	**Contract: C-05 Validation on FastAPI** Write the contract for the instant check of an override or yard adjustment on the internal AI service, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	DEC-03 DEC-18	—

AI Team · 25 tasks

ID	Task	Needs from own team	Needs from other teams
CT-14	**Contract: C-06 Completion message** Write the contract for the "finished" message every worker sends, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	DEC-19	DEC-01 (Backend)
CT-15	**Contract: AI result tables and write ownership** Write the contract for which team creates and which team completes each AI result table, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	—	DEC-02 (Backend)
CT-16	**Contract: FastAPI internal API** Write the contract for the internal AI service: its endpoints, who may call it and how, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	DEC-10	—
CT-18	**Contract: AI seed data and port_setting keys** Write the contract for the AI reference data and settings the Backend loads, as described in the Contracts file. The AI Team writes it and the Backend review and approve it. Done when it is approved and published in the contracts folder.	DEC-12 DEC-23	—

Backend Team · 30 tasks

ID	Task	Needs from own team	Needs from other teams
P1-A1	**Domain modules and Docker Compose (PHP, NGINX, MySQL 8, Redis)** Set up the Backend project so every team member can run the full stack (application, web server, database and Redis) on their own machine with one command. Split the code into the business modules agreed in the architecture. Done when a new member can clone the repository and see the application running.	—	—
P1-A2	**API base: one address prefix, standard error format with stable codes, paging, message keys, UTC** Build the shared base every endpoint will use: one address prefix for the API, one error format with stable error codes and readable messages, the same paging style for every list, and all times stored and returned in UTC. Done when a sample endpoint shows each of these and follows the conventions contract (CT-01).	P1-A1 CT-01	—
P1-A3	**CI on every push: Pint, Larastan, Pest with MySQL and Redis** Make sure every push is checked automatically: code style, static checks and tests running against a real database and Redis. A broken push must be visible to the team straight away. Done when a failing test blocks a pull request.	P1-A1	—

Backend Team · 30 tasks

ID	Task	Needs from own team	Needs from other teams
P1-A4	**Publish the OpenAPI file and keep it current every sprint** Publish the API description file that the Frontend and Mobile teams build against, and keep it updated in the same pull request whenever an endpoint changes. Done when the file is in the agreed place and matches what the server really does.	P1-A2 DEC-13	—
P1-B1	**Load PortSync SQL and DDL fixes; generate draft migrations** Load the agreed port database design, apply the known corrections, and turn it into database migrations. Done when the migrations create every table of the design on an empty database.	P1-A1	—
P1-B2	**Review migrations: access, marine, terminals, storage, yard, equipment** Review the generated migrations for access, marine resources, terminals, storage, yard and equipment against the database design: names, types, keys and relations. Fix anything that does not match. Done when the reviewed group is merged.	P1-B1	—
P1-B3	**Review migrations: request, BAPLIE, Part 1, Part 2, plan, execution, learning** Review the generated migrations for requests, BAPLIE, AI Part 1 and Part 2 results, plans, execution and learning, including the tables the AI writes to. Done when the reviewed group is merged and the AI Team agrees with the tables it uses.	P1-B1	—
P1-B5	**All Eloquent models and relations; one PHP enum per status column** Create the data model for every table with its relations, and one fixed list of allowed values for every status column so all code uses the same status names. Done when every table can be read and written through the model layer.	P1-B2 P1-B3	—
P1-C1	**Web sign-in (Sanctum SPA, CSRF, 422/401/403); profile with roles and terminal scope** Let office staff sign in on the web and get back their profile, role and terminal scope, with clear answers for missing fields, wrong credentials and disabled accounts. Done when the web sign-in works end to end with the agreed identifier (DEC-35).	P1-B2 DEC-35	—
P1-C2	**must_change_password gate** Force a user who has a one-time password to change it before doing anything else. Done when such a user cannot reach any other function until the password is changed.	P1-C1	—
P1-B4	**Real CHECK constraints, generated columns and ENUMs** Add the database rules that protect data quality: allowed values, checks between columns and calculated columns, exactly as the design states. Done when invalid rows are refused by the database itself.	P1-B2 P1-B3	—
DEC-01	**Close DEC-01: AI completion signal** Laravel must learn that a worker finished so it can send SSE and notifications within 5 seconds of the commit. No mechanism is documented. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

Backend Team · 30 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-02	**Close DEC-02: Who writes status transitions, history, audit and run rows** Request status (SM-4), its history and its audit need a single writer. solver_run.solver_version is mandatory and only the worker knows it. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-05	**Close DEC-05: SSE source for the dashboard** The AI documents do not say what feeds the SSE. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-08	**Close DEC-08: Database read source and worker privileges** Each worker must read port state and write only its own results. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-13	**Close DEC-13: Who keeps the OpenAPI file current** Frontend and Mobile build against the OpenAPI file, so it must stay up to date. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-35	**Close DEC-35: Sign-in identifier** Screens: email on the web, username on the tablet. The FRD creates users with name, username and phone, with no email. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
CT-01	**Contract: Port API conventions (errors, pagination, headers)** Write the contract for the common rules of the port API: address prefix, error format and codes, paging, headers and the record version, as described in the Contracts file. The Backend Team writes it and the Frontend, Mobile and QA review and approve it. Done when it is approved and published in the contracts folder.	—	—
CT-02	**Contract: Port API OpenAPI file** Write the contract for the API description file and the rule that it changes in the same pull request as the endpoint, as described in the Contracts file. The Backend Team writes it and the Frontend, Mobile and QA review and approve it. Done when it is approved and published in the contracts folder.	CT-01 DEC-13	—
CT-03	**Contract: Web sign-in and session** Write the contract for how office staff sign in on the web, change a one-time password and sign out, as described in the Contracts file. The Backend Team writes it and the Frontend review and approve it. Done when it is approved and published in the contracts folder.	DEC-35	—

Backend Team · 30 tasks

ID	Task	Needs from own team	Needs from other teams
CT-04	**Contract: Mobile sign-in and system info** Write the contract for how the tablet learns its port, signs in and is told to update, as described in the Contracts file. The Backend Team writes it and the Mobile review and approve it. Done when it is approved and published in the contracts folder.	DEC-35	—
CT-05	**Contract: Web SSE events** Write the contract for the list of live events the web receives and what each one carries, as described in the Contracts file. The Backend Team writes it and the Frontend review and approve it. Done when it is approved and published in the contracts folder.	DEC-05	—
CT-06	**Contract: Mobile WebSocket events** Write the contract for the live events the tablet receives on its private channel, as described in the Contracts file. The Backend Team writes it and the Mobile review and approve it. Done when it is approved and published in the contracts folder.	—	—
CT-07	**Contract: Mobile sync (Outbox events)** Write the contract for how the tablet sends offline updates and fetches changes, and the types of update it can send, as described in the Contracts file. The Backend and Mobile Team writes it and the Mobile and Backend review and approve it. Done when it is approved and published in the contracts folder.	—	—
CT-17	**Contract: Database schema export (schema.sql)** Write the contract for the database schema export the AI tests are built on, and how schema changes are requested, as described in the Contracts file. The Backend Team writes it and the AI review and approve it. Done when it is approved and published in the contracts folder.	P1-B5 DEC-08	—
CT-19	**Contract: Mock worker behaviour** Write the contract for how the stand-in AI workers behave and how to choose a scenario, as described in the Contracts file. The Backend Team writes it and the Frontend, Mobile and QA review and approve it. Done when it is approved and published in the contracts folder.	—	CT-08 (AI)
CT-20	**Contract: AI result read endpoints** Write the contract for the endpoints that show AI results on the screens, as described in the Contracts file. The Backend Team writes it and the Frontend review and approve it. Done when it is approved and published in the contracts folder.	—	—
CT-21	**Contract: Shared status vocabulary** Write the contract for the single list of every status value, as described in the Contracts file. The Backend Team writes it and the all teams review and approve it. Done when it is approved and published in the contracts folder.	P1-B5	—

Backend Team · 30 tasks

ID	Task	Needs from own team	Needs from other teams
CT-23	**Contract: API readiness pack template** Write the contract for what the API readiness pack contains at every Feature Readiness Review, as described in the Contracts file. The Backend Team writes it and the Frontend, Mobile and QA review and approve it. Done when it is approved and published in the contracts folder.	CT-02	—
BE-PK1	**API readiness pack for FRR 2** Prepare the API readiness pack for the features handed off at FRR 2: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 2.	CT-23	—

UI/UX Team · 7 tasks

ID	Task	Needs from own team	Needs from other teams
UX-W1	**Final UI: shells, sign-in, shared and tablet base pages (P-16, P-17, P-19, P-20, P-52, P-53, P-56, P-57, P-58)** Finalise the shells and the shared and tablet base screens from the Screens Specification, with all their states. Done when the Frontend and Mobile teams accept them.	—	—
UX-W2	**Final UI: Port Admin and safety pages (P-23 to P-29, P-59, P-62)** Finalise the Port Admin and safety screens with all their states. Done when the Frontend Team accepts them.	—	—
UX-W3	**Final UI: intake and department pages (P-31, P-33 to P-36, P-43, P-45, P-46, P-48, P-49)** Finalise the intake and department screens with all their states. Done when the Frontend Team accepts them.	—	—
UX-W4	**Final UI: AI gates and field pages (P-37, P-38, P-39, P-40, P-54, P-55)** Finalise the AI gate screens and the field task screens with all their states. Done when the Frontend and Mobile teams accept them.	—	—
UX-W5	**Final UI: dispatch and monitoring pages (P-44, P-47, P-41, P-42, P-18, P-21)** Finalise the dispatch and monitoring screens with all their states. Done when the Frontend Team accepts them.	—	—
UX-W6	**Final UI: yard and governance pages (P-50, P-51, P-30, P-17 widgets)** Finalise the yard and governance screens and the final dashboard widgets. Done when the Frontend Team accepts them.	—	—
CT-22	**Contract: Screen design and tokens hand-off** Write the contract for how screens, their states and the design tokens are handed over, as described in the Contracts file. The UI/UX Team writes it and the Frontend and Mobile review and approve it. Done when it is approved and published in the contracts folder.	UX-W1	—

Frontend Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
FE-001	**Approve the frontend stack and repository layout (ADR)** Choose and record the web technology stack and how the repository is organised, so everyone follows the same structure. Done when the decision record is merged.	—	—
FE-006	**Fix language, time zone, accessibility and browser scope** Fix the scope of the web app: display language, time zone (DEC-45), accessibility level and the browsers and screen sizes to support (DEC-46). Done when the choices are written and shared with QA.	DEC-45 DEC-46	—
FE-010	**Scaffold the frontend workspace** Create the web project so every member can run it locally and build it. Done when a new member can clone and run it.	FE-001 FE-006	—
FE-011	**Runtime and build configuration (future-work flags)** Set up the settings for running and building the app in each environment. Done when the same build runs on the dev and demo servers with only settings changed.	FE-010	—
FE-012	**Continuous integration pipeline** Check every push automatically: style, tests and build. Done when a broken push blocks a pull request.	FE-010	—
FE-015	**Design tokens and colour contrast check** Turn the approved brand colours, fonts and spacing (CT-22) into reusable design tokens and check colour contrast. Done when pages use only the tokens.	FE-010	UX-W1 (UI/UX)
FE-022	**Role navigation configuration** Configure the side menu per role so each user sees only their pages. Done when every role sees the right menu.	FE-010	—
DEC-45	**Close DEC-45: Display time zone** The database is UTC and shifts are local time. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-46	**Close DEC-46: Supported browsers and viewports** The test scope must be defined. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
FE-S01	**Static pages: landing, loading, empty, error and not found (no UI design; Frontend decides)** Build the pages that need no API and no design hand-off: the landing page, loading, empty and error screens and the not-found page. The Frontend Team decides their look within the brand. Done when the pages are reachable and on brand.	FE-010	—

Mobile Team · 5 tasks

ID	Task	Needs from own team	Needs from other teams
MB-F1	**Setup: Feature-First structure, get_it, go_router role redirect, brand colours and fonts, AR/EN with RTL, responsive helper** Create the tablet app with its structure, navigation by role, brand colours and fonts, Arabic and English with right-to-left support, and layouts that also work on phones. Done when the empty app runs in both languages.	—	—
MB-F2	**Network layer: Dio client, auth / app-version / device-id / idempotency interceptors, error mapping (401, 403, 409, 422, 426), Root CA trust, ping** Build one way for the app to talk to the server that adds the needed identity and device information and turns every error into a clear message (CT-04). Done when every error type shows the right message.	MB-F1	—
MB-P57	**P-57 App Update Required (static screen, blocks actions, keeps Outbox)** Build P-57, which blocks the app when its version is too old and keeps unsent work safe. Done when an old version cannot be used.	MB-F1	UX-W1 (UI/UX)
MB-P20	**P-20 Access Denied (tablet variant)** Build the tablet version of P-20 Access Denied. Done when refused actions lead to it.	MB-F1	UX-W1 (UI/UX)
MB-S01	**Static screens: splash, loading, empty and error (no UI design; Mobile decides)** Build the screens that need no API and no design hand-off: splash, loading, empty and error. The Mobile Team decides their look within the brand. Done when the screens are used across the app.	MB-F1	—

QA Team · 5 tasks

ID	Task	Needs from own team	Needs from other teams
QA-01	**Test strategy: levels, environments, evidence rules, bug workflow on release branches** Write how the project is tested: test levels, environments, what counts as evidence, and how defects are reported and fixed on the release branches (CT-26). Done when all teams agree.	CT-26	—
QA-AI1	**Draft the thresholds register inputs and the manual-planning baseline procedure with domain contacts** Gather the inputs the AI needs for its thresholds and describe how manual planning is done today so the AI can be compared with it. Done when the AI Team accepts the inputs.	—	—
CT-24	**Contract: BAPLIE test corpus format** Write the contract for the layout of the BAPLIE test files and their expected results, as described in the Contracts file. The QA Team writes it and the AI review and approve it. Done when it is approved and published in the contracts folder.	—	—

QA Team · 5 tasks

ID	Task	Needs from own team	Needs from other teams
CT-25	**Contract: Reference data and thresholds format** Write the contract for the layout of the reference data, with the source of every value, as described in the Contracts file. The QA Team writes it and the AI review and approve it. Done when it is approved and published in the contracts folder.	—	—
CT-26	**Contract: QA report and defect format** Write the contract for the release test report and the defect report, as described in the Contracts file. The QA Team writes it and the all teams review and approve it. Done when it is approved and published in the contracts folder.	—	—

M2 · Secure access

Sprint 2 — Sign-in, access control and platform core

46 tasks
Release
v0.2.0

Goal. Deliver sign-in, roles and audit on the Backend and hand the web sign-in to the Frontend Team. The AI Team builds the shared worker engine, the data access layer and the internal AI service.

Handed off at the FRR

Hand-off	Approved items	To
FT-02 Web sign-in	P1-C1 P1-C2 UX-W1	Frontend
Tablet base screens	UX-W1	Mobile

Done when

- Web sign-in, password change and sign-out work against the real API
- Roles, terminal scope and audit work
- The AI worker engine completes a test job
- The internal AI service is running

Leads Council · 4 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR2	**Sprint 2 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 2, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK1 (Backend) CT-22 (UI/UX)
PL-REL2	**Sprint 2 review and release: develop → release → main, tag v0.2.0, back-merge, PortHub, demo deploy** At the end of Sprint 2, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v0.2.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT2 (QA)
DEC-42	**Close DEC-42: Staff pages on the tablet and a provisioning screen** They are not in the requirements. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-48	**Close DEC-48: Field roles on the web** What happens when a pilot, tug master or CT Supervisor signs in on the web. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

AI Team · 9 tasks

ID	Task	Needs from own team	Needs from other teams
AI-008	**Thresholds register and evaluation baseline (drafted with the QA Team)** Write the register of all numbers the AI depends on (drift limits, deviation significance, data sufficiency, tide tolerance, file size classes) and the baseline used to judge the AI, based on the decisions in DEC-30. Done when every number has a value and an owner.	DEC-30	QA-AI1 (QA)
AI-009	**AI seed data specification: constraint types, objective terms, rationale templates, model keys** Specify the reference data the AI needs in the database: constraint types, goal terms and weights, explanation templates and model names (CT-18). Done when the Backend can load it without questions.	DEC-23	—
AI-011	**Shared worker runtime: queue consumer, idempotent claim, retries, stale sweep, logging** Build the common engine every worker runs on: pick up work from its queue, make sure the same work is never done twice, retry on failure, mark stuck runs, log everything, and send the "finished" message (CT-14). Done when a test job travels from the queue to the database and back.	CT-08 AI-010 DEC-04	DEC-01 (Backend)
AI-012	**Configuration management (environment + port_setting)** Read AI settings from the environment and from the port settings table, so values change without code. Done when every worker reads its settings this way.	AI-010 DEC-12	—
AI-013	**Data-access layer on a schema built from the PortSync DDL; per-worker accounts** Build the layer the workers use to read port data and write results, on a database built from the Backend migrations (CT-17), with one account per worker. Done when workers read and write only their own tables.	AI-010 CT-08	DEC-08 (Backend) CT-17 (Backend)
AI-054	**Tug-requirement computation from the approved formula** Calculate how many tugs each vessel needs from the approved rule (DEC-20). Done when the rule gives the expected answer on known cases and can be replaced without changing code.	DEC-20	—
DEC-20	**Close DEC-20: Tug-count formula** The number of tugs per vessel (from GT, DWT and tug power) is not defined, and this blocks CP-SAT. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-30	**Close DEC-30: Numeric thresholds register** Drift, deviation significance, data sufficiency, tide tolerance, BAPLIE size classes, coverage. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	QA-AI1 (QA)
AI-F01	**FastAPI service: structure and internal auth (X-Service-Token)** Set up the internal AI web service that only the Backend can call, protected by a service key (DEC-10, CT-16). Done when the Backend can reach it and nobody else can.	AI-010 DEC-10 CT-16	—

Backend Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
P1-B6	**Seeders: reference rows and a Damietta-scale demo port (8 terminals, 36 berths, pilots, tugs, shifts)** Fill the database with reference data and a realistic demo port at the size of the reference port (terminals, berths, pilots, tugs, shifts) so every team can test with believable data. Done when a fresh environment starts with the demo port loaded.	P1-B5	CT-18 (AI)
P1-C3	**Mobile sign-in: device id, platform, app version, 426, 30-day sliding token, user_device** Let field users sign in from the tablet. The device is registered, the app version is checked (old versions are told to update), and the user gets a session that stays valid while in use. Done when a tablet can sign in and an outdated app is refused.	P1-B2 DEC-35	—
P1-C4	**Logout: end web session; revoke mobile token** Let users sign out: the web session ends, and the tablet session is cancelled on the server so the device can no longer act for that user. Done when a signed-out session is refused everywhere.	P1-C1 P1-C3	—
P1-C5	**System info: port id, port name and minimum app version** Give the tablet a simple public way to learn which port it is connected to and the minimum app version it needs. Done when the app can show the port name on first launch.	P1-B2	—
P2-D1	**Role policies for the 12 roles, terminal-scope middleware (403), field users see own tasks** Apply the permissions of the 12 roles: each role reaches only its own functions, terminal-scoped users see only their terminal, and field users see only their own tasks. Done when a test for every role passes and out-of-scope requests are refused.	P1-C1	—
P2-D2	**Write chain: auth, password gate, scope, app version, version token, rules** Make every change pass the same checks in the same order: signed in, password changed, allowed scope, app version, record not changed by someone else, business rules. Done when each check gives its own clear answer.	P2-D1	—
P2-D3	**Audit service: write + audit row in one transaction; refused writes as REJECTED_ATTEMPT** Record every change together with who made it, in the same step as the change itself, and also record refused attempts. Done when no change can be saved without its audit entry.	P1-B5	—
P2-D4	**Version token on updated_at; stale write returns 409** Stop two people from overwriting each other: every record carries a version, and saving with an old version is refused with the current data. Done when a stale save is refused with a conflict answer.	P2-D2	—

Backend Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
P2-E1	**Create user with one-time password; deactivate under a row lock (409 self / last Port Admin); first Port Admin seeder** Let the Port Admin create users with a one-time password and deactivate them. Nobody can deactivate themselves or the last Port Admin, even if two admins try at the same time. Done when these rules hold under parallel attempts.	P2-D1 P2-D3	—
P2-E2	**Assign role with unit check; end role with active_to** Let the Port Admin give a user a role (checked against the user's unit) and end it, keeping the role history (DEC-36). Done when role changes are visible in the user's history.	P2-E1 DEC-36	—
P2-E3	**List a user's devices; revoke one device or all tokens** Show each user's registered devices and let the Port Admin cancel one device or all of them, for example when a tablet is lost. Done when a cancelled device can no longer act.	P2-E1	—
P2-F8	**Audit log browsing; master-data history as a filter** Let the Port Admin browse the audit log and filter it, including the change history of master data. Done when any change can be traced to who, what and when.	P2-D3	—
DEC-36	**Close DEC-36: One role per user vs role history** The screens say one role; the Backend assigns and ends roles with dates. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-47	**Close DEC-47: Backend environment for development and testing** Frontend, Mobile and QA need a server with the latest develop and seed data. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
BE-PK2	**API readiness pack for FRR 3** Prepare the API readiness pack for the features handed off at FRR 3: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 3.	BE-PK1	—

UI/UX Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
UX-QA2	**Design QA of the screens built in Sprint 2** Compare the screens built in Sprint 2 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV2	**Revise designs from the Sprint 2 review and design QA findings** Update the designs based on the Sprint 2 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA2	—

Frontend Team · 11 tasks

ID	Task	Needs from own team	Needs from other teams
FE-016	**Core component library (static, loading, empty and error states included)** Build the shared building blocks every page uses (tables, forms, dialogs, filters, states) from the approved designs. Done when pages can be assembled from the library.	FE-015	CT-22 (UI/UX)
FE-017	**Status badge registry** Build one place that turns every status into its badge colour and label, using the shared status list (CT-21). Done when every status shows the same badge everywhere.	FE-015	CT-21 (Backend)
FE-030	**HTTP client and error normalisation** Build one way to call the API that handles sign-in, errors and conflict answers the same way everywhere (CT-01). Done when every error shows a clear message.	FE-010	CT-01 (Backend)
FE-031	**Resource composables and cache** Build reusable data loaders with caching so pages load quickly and stay consistent. Done when pages share data without reloading it.	FE-030	—
FE-032	**Version-token handling and 409 conflict UX** Handle "someone else changed this record" answers: show the current data and let the user retry. Done when a conflict never loses the user's work silently.	FE-031	—
FE-034	**MSW mock layer and fixtures (test use only)** Build a mock API used only in tests, so tests do not depend on the server. Done when the test suite runs without a server.	FE-030	—
FE-020	**Port app shell** Build the main frame of the app: header, side menu and content area. Done when every page opens inside the frame.	FE-016 FE-022	—
FE-021	**Router, guards, P-20 Access Denied and not-found page** Build navigation and access rules: pages a role may not open lead to Access Denied (P-20), and unknown addresses lead to Not Found (DEC-48). Done when no role can reach a page outside its rights.	FE-010 FE-027	DEC-48 (Leads)
FE-025	**P-16 Port Login** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-16 lets office staff sign in with the agreed identifier (CT-03, DEC-35). Done when sign-in works against the real API.	FE-030 FE-016	P1-C1 (Backend) UX-W1 (UI/UX) CT-03 (Backend)
FE-026	**Forced password change** Force users with a one-time password to set a new one before continuing. Done when they cannot skip it.	FE-025	P1-C2 (Backend)
FE-027	**Session lifecycle, logout and multi-tab handling** Handle the session: sign out, expired sessions and several open tabs. Done when an expired session always returns to sign-in.	FE-030	—

Mobile Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
MB-F3	**Local storage: encrypted Hive boxes, secure key, TokenStorage, Outbox and cache models, meta box** Store data securely on the device: the session, cached tasks and notifications, and the queue of updates waiting to be sent. Done when data survives a restart and stays encrypted.	MB-F1	—
MB-F6	**App shell and shared states: bottom navigation, connection/sync bar, event state chip, loading/empty/error, routing to P-20 / P-57 / P-52** Build the app frame and the shared states: bottom navigation, connection and sync bar, and the automatic moves to P-20, P-57 or sign-in. Done when every screen opens inside the frame.	MB-F1	UX-W1 (UI/UX) CT-22 (UI/UX)

QA Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
QA-AI2	**Author BAPLIE corpus cases and expected-result files (valid, invalid, quarantine, unreadable)** Write the BAPLIE test cases and their expected results in the agreed format (CT-24). Done when the AI Team accepts the corpus.	CT-24	—
QA-TC2	**Acceptance test cases for the features handed off at FRR 2** Write the acceptance tests for every feature handed over at FRR 2, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT2	**Sprint 2 release testing on the release branches and sign-off report** Test the Sprint 2 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC2	—

M3 · Master data and AI bridge

Sprint 3 — Master data and the AI bridge

46 tasks
Release v0.3.0

Goal. Finish every master-data API and start the request flow and the link to the AI. The AI Team starts Part 2 and the BAPLIE parser in parallel. The Frontend Team builds user management and audit; the Mobile Team builds tablet sign-in.

Handed off at the FRR

Hand-off	Approved items	To
FT-02 Tablet sign-in	P1-C3 P1-C4 P1-C5 UX-W1	Mobile
FT-03 Users, devices and audit	P2-E1 P2-E2 P2-E3 P2-F8 UX-W2	Frontend

Done when

- User and audit pages pass QA on the release branch
- Tablet sign-in works with device registration
- All master-data APIs are merged with examples and demo data
- The BAPLIE parser reads the test files and the Part 2 loader runs on synthetic data

Leads Council · 2 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR3	**Sprint 3 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 3, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK2 (Backend)
PL-REL3	**Sprint 3 review and release: develop → release → main, tag v0.3.0, back-merge, PortHub, demo deploy** At the end of Sprint 3, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v0.3.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT3 (QA)

AI Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
AI-015	**Test harness and synthetic reference-port scenario generator** Build a test toolkit and a generator of realistic, clearly labelled synthetic port scenarios at the size of the reference port. Done when any AI part can be tested on generated scenarios.	AI-013	—
AI-050	**Port-state loader for Part 2 (repeatable-read snapshot)** Load a consistent snapshot of the port (berths, tugs, pilots, bookings, requests) for Part 2 to plan on (DEC-06, DEC-32). Done when the snapshot matches the database at one moment.	AI-013 DEC-06 DEC-32	—

AI Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
AI-052	**B3 service time Phase 1 formula** Estimate how long a vessel stays at berth from the terminal speed and the number of moves (DEC-28). Done when estimates match hand calculations.	AI-050 DEC-28	—
AI-053	**B4 delay-risk Phase 1 rule (NULL until a rule is approved)** Estimate the delay risk of a plan using the approved simple rule, or leave it empty if no rule is approved (DEC-21). Done when every option carries a risk value or a clear "not assessed".	DEC-21 AI-050	—
AI-044	**A4 threshold rules with strictest-wins scope resolution (built first on synthetic series)** Apply the safety limits (wind, waves, visibility, tide) with the strictest rule winning, built first on synthetic weather data so Part 2 is not blocked (DEC-34). Done when blocked periods match the limits on test cases.	AI-009 AI-014 DEC-34 DEC-25	—
AI-023	**BAPLIE D.17A test corpus (authored with the QA Team)** Assemble the BAPLIE test files with their expected results, together with the QA Team (CT-24). Done when the corpus covers valid, wrong-vessel, invalid and unreadable files.	—	QA-AI2 (QA)
AI-030	**BAPLIE D.17A segment parser and SMDG mapping** Read BAPLIE files in the standard format and map their content to vessels, containers and positions (DEC-17). Done when every corpus file gives its expected result.	AI-011 AI-023 DEC-17 CT-12	—
AI-065	**Yard placement method decision (Option A heuristic), constraints, confidence** Fix the method for yard placement (rule-based), its constraints and how confidence is measured (DEC-29). Done when the method is written down and agreed with the CT Planner role.	DEC-29	—
DEC-21	**Close DEC-21: Phase 1 delay-risk rule** B4 needs a rule to estimate delay probability before any history exists. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-25	**Close DEC-25: Uncertainty margin and window confidence** A6 (margin) is future work, so MVP behaviour must be defined. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-28	**Close DEC-28: Definition of moves** Service time (B3) needs the number of moves. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

AI Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-29	**Close DEC-29: Yard recommendation method, objective and confidence** The B8 method is not defined. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-32	**Close DEC-32: Availability semantics for pilots and tug masters** B2 and B5 need a definition of "available". Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-34	**Close DEC-34: Building A4 rules on synthetic data** The documented critical path is AI-040 → 041 → 043 → 044, but Part 2 needs AI-044 early. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
AI-F02	**FastAPI: health, workers and runs endpoints** Expose the health of the service and of each worker, and the status of any run, for diagnosis. Done when the Backend can see if a worker is down.	AI-F01 AI-011	—

Backend Team · 13 tasks

ID	Task	Needs from own team	Needs from other teams
P2-F1	**Port profile, navigation channel (chart datum), waypoints, typed port_setting editor** Manage the port profile, the navigation channel with its reference depth level, the channel waypoints, and the port settings that change without code. Done when all of these can be read and edited with validation.	P2-D2	—
P2-F2	**Terminals, berths (max LOA within length), berth_terminal (shared berths allowed)** Manage terminals and berths. A berth's maximum vessel length must fit its length, and a berth can serve one or more terminals. Done when invalid berths are refused and shared berths work.	P2-D2	—
P2-F5	**Safety constraints (no overlap); tide constituent set activation with datum check** Manage safety constraints without overlaps, and activate the tide data set only when its reference level matches the channel and berths (DEC-38). Done when overlapping constraints and mismatched levels are refused.	P2-D2	DEC-38 (UI/UX)
P2-F3	**Tugs, tug masters (one per tug), pilots, rank limits, shifts, pilot and tug master schedules** Manage tugs, tug masters (one per tug), pilots, their ranks and the vessel sizes each rank may handle, port shifts, and pilot and tug master schedules. Done when every resource the AI needs can be maintained.	P2-D2	—

Backend Team · 13 tasks

ID	Task	Needs from own team	Needs from other teams
P2-F4	**CT equipment and capabilities; yard zones, blocks and slots (reefer, DG, OOG)** Manage container terminal equipment and what it can do, and the yard structure down to individual slots, including reefer, dangerous goods and oversized areas. Done when the full yard of a terminal can be entered.	P2-D2	—
P2-F6	**Terminal productivity profiles (no overlapping periods)** Manage how fast each terminal works over time, without overlapping periods. The AI uses this to estimate how long a vessel stays at berth. Done when overlapping periods are refused.	P2-F2	—
P2-F7	**ResourceChanged domain event on berth, tug, pilot, equipment changes** Announce internally whenever a berth, tug, pilot or equipment changes state, so pending recommendations that use it can be refreshed. Done when a resource change reliably triggers the event.	P2-F2 P2-F3	—
P2-F9	**Seed AI port_setting keys and AI reference rows from the AI-009 spec** Load the AI settings and AI reference data agreed in the AI seed contract (CT-18). Done when the AI workers find every setting they need.	—	AI-009 (AI) CT-18 (AI)
P3-G1	**Create request: vessel by IMO (check digit), per-call particulars, cargo lines** Let staff create a berthing request: find the vessel by its IMO number (with a validity check), enter the call details and the cargo lines. Done when a complete request can be created and invalid input is refused clearly.	P2-F2	—
P3-G2	**SM-4 transition service: status, history and audit in one transaction** Make every status change of a request follow the agreed life cycle, and save the status, its history and its audit together. Done when no illegal status change is possible.	P3-G1 P2-D3	—
P3-I1	**Job publisher: JSON envelope v1 (RPUSH), builders for C-01 to C-05** Send work to the AI workers using the agreed message format (CT-08 to CT-12), one queue per worker. Done when each message type reaches the right queue in the agreed shape.	—	CT-08 (AI) DEC-04 (AI) CT-09 (AI) CT-10 (AI) CT-11 (AI) CT-12 (AI)
DEC-11	**Close DEC-11: Redis persistence** If Redis goes down, queued jobs may be lost. Persistence is not documented. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
BE-PK3	**API readiness pack for FRR 4** Prepare the API readiness pack for the features handed off at FRR 4: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 4.	BE-PK2	—

UI/UX Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-38	**Close DEC-38: Where to activate the tide constituent set** Activation with the datum check can live on P-59 or P-26. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
UX-QA3	**Design QA of the screens built in Sprint 3** Compare the screens built in Sprint 3 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV3	**Revise designs from the Sprint 3 review and design QA findings** Update the designs based on the Sprint 3 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA3	—

Frontend Team · 8 tasks

ID	Task	Needs from own team	Needs from other teams
FE-018	**Icon set** Add the icon set used across the pages. Done when every icon in the designs is available.	FE-015	—
FE-100	**Unit and component test baseline** Set up unit and component tests with a few first tests as an example. Done when tests run in the pipeline.	FE-010 FE-034	—
FE-117	**Developer documentation and decision log** Write the developer guide and keep a log of frontend decisions. Done when a new member can start from it.	FE-010	—
FE-035	**Types generated from OpenAPI** Generate the data types from the API description file so the app always matches the API (CT-02). Done when an API change shows up as a build error.	FE-030	CT-02 (Backend)
FE-045	**P-23 User Management and P-24 User Details** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-23 lists users with filters; P-24 shows one user, their role and terminal scope, and lets the Port Admin create, edit and deactivate. Done when the Port Admin can manage users end to end.	FE-021 FE-032	P2-E1 (Backend) P2-E2 (Backend) UX-W2 (UI/UX)
FE-046	**P-25 Registered Devices** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-25 lists registered devices and lets the Port Admin cancel one or all. Done when a cancelled device disappears from the active list.	FE-045	P2-E3 (Backend)

Frontend Team · 8 tasks

ID	Task	Needs from own team	Needs from other teams
FE-050	**P-29 Port Audit and Change History** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-29 lets the Port Admin search the change history and see before and after values. Done when any change can be found.	FE-021 FE-031	P2-F8 (Backend) UX-W2 (UI/UX)
FE-040	**P-17 Home Dashboard (shell and first widgets; widgets grow each sprint)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-17 is the start page; build the frame and the first widgets now, and add each role's widgets as their features arrive. Done when every role lands on a useful dashboard.	FE-020 FE-031	—

Mobile Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
MB-P53	**P-53 Port Identification: GET /system/info, min app version check, offline fallback** Build P-53, which shows which port the tablet is connected to on first launch and checks the app version (CT-04). Done when the port name appears and offline launch uses the last known port.	MB-F2	P1-C5 (Backend) CT-04 (Backend)
MB-P52	**P-52 Tablet Login: device data, token, forced password change, same-user Outbox rule** Build P-52 so field users can sign in, change a one-time password, and sign in again after a session expires without losing unsent work (CT-04). Done when sign-in works and another user can never send someone else's updates.	MB-F2 MB-F3	P1-C3 (Backend) CT-04 (Backend)
MB-P58	**P-58 Settings and Session: language switch, logout online/offline, keep pending events** Build P-58 with account, port and device information, the language switch and sign-out that warns about unsent work. Done when sign-out keeps unsent work safe.	MB-P52	P1-C4 (Backend)

QA Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
QA-TC3	**Acceptance test cases for the features handed off at FRR 3** Write the acceptance tests for every feature handed over at FRR 3, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT3	**Sprint 3 release testing on the release branches and sign-off report** Test the Sprint 3 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC3	—

M4 · Requests, BAPLIE and the engine

Sprint 4 — Requests, BAPLIE and the Part 2 engine

42 tasks
Release
v0.4.0

Goal. Complete the request and BAPLIE APIs, the live update channel and the stand-in AI workers so no team waits for a real worker. The AI Team builds the optimisation model and the BAPLIE save logic. The Frontend Team builds the port master-data pages.

Handed off at the FRR

Hand-off	Approved items	To
FT-04 Port master data	P2-F1 P2-F2 P2-F5 UX-W2	Frontend
FT-05 Pilots	P2-F3 UX-W3	Frontend

Done when

- Stand-in AI workers answer every queue in the contract shape
- Live updates are filtered by role and terminal
- The optimisation model returns ranked options at port scale
- Port master-data pages pass QA

Leads Council · 2 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR4	**Sprint 4 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 4, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK3 (Backend)
PL-REL4	**Sprint 4 review and release: develop → release → main, tag v0.4.0, back-merge, PortHub, demo deploy** At the end of Sprint 4, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v0.4.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT4 (QA)

AI Team · 11 tasks

ID	Task	Needs from own team	Needs from other teams
AI-051	**B2 constraint filter (also applied to FIFO output)** Remove every option that breaks a hard rule, including options produced by the fallback method. Done when no emitted option ever breaks a hard rule in random tests.	AI-050 AI-044	—
AI-055	**B5 CP-SAT allocation model with ranking (10 s limit)** Build the optimisation model that assigns berths, tugs and pilots across the port and ranks the options, within the 10-second limit (DEC-22, DEC-23). Done when it returns ranked, valid options on reference-size scenarios in time.	AI-051 AI-052 AI-054 DEC-06 AI-009	—

AI Team · 11 tasks

ID	Task	Needs from own team	Needs from other teams
AI-031	**Shared BAPLIE validation rule set and diagnostics** Write the BAPLIE checking rules once so they serve both the parser and staff corrections, with clear problem messages. Done when the same file gives the same diagnostics in both paths.	AI-030 DEC-03	—
AI-032	**All-or-nothing BAPLIE commit, quarantine and failure handling** Save a parsed file all at once or not at all, hold suspicious files for review, and report failures clearly. Done when no failure leaves half-saved data.	AI-031 AI-013	—
AI-020	**Select weather and marine providers and build adapters (recorded payloads first)** Choose the weather and sea data providers and connect to them, starting with recorded sample data (DEC-26). Done when both providers return every needed variable in our units.	AI-012 AI-014 DEC-26	—
AI-021	**Acquire and validate tidal constants and datum** Obtain and check the tide constants and the reference level for the port (DEC-27). Done when predicted tides match the published reference within the agreed tolerance.	AI-014 DEC-27	—
AI-025	**Data-quality checks library** Build shared checks for data quality (ranges, gaps, freshness). Done when every incoming data set is checked before use.	AI-014	—
DEC-22	**Close DEC-22: Financial-impact data in ranking** The documented order is time first, then financial impact, but there is no cost data. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-24	**Close DEC-24: Part 1 settings: horizon, time step, minimum window, slot** The window computation needs fixed values set by the Marine Safety Officer and staff. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-26	**Close DEC-26: Weather and marine providers** A primary and a fallback are needed covering every A4 variable (wind, visibility, waves, current, sea level). Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-27	**Close DEC-27: Tidal constants source, datum and reference** A2 needs published constants for the reference port and a single datum shared with the depths. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

Backend Team · 16 tasks

ID	Task	Needs from own team	Needs from other teams
P3-G3	**Edit, cancel, preliminary approval, return, resubmit, reject** Support every review decision on a request: edit, cancel, preliminary approval, return for revision, resubmit and reject, each allowed only in the right status. Done when each decision moves the request correctly.	P3-G2	—
P3-G4	**List and search by status, vessel, terminal and ETA** Let staff list and search requests by status, vessel, terminal and arrival time. Done when the list answers these filters quickly.	P3-G1	—
P3-H1	**BAPLIE upload to the shared volume, SHA-256, duplicate rule (409), STAGED/PENDING** Let staff upload a BAPLIE file. Store it safely, refuse an exact duplicate for the same request, and mark it as waiting for parsing. Done when uploads and duplicate refusals work.	P3-G1	—
P3-H2	**Auto-match by IMO and ETA within 24 h (422 on 0 or many)** Link an uploaded file to its request automatically using the vessel and the expected arrival time, and explain clearly when no request or more than one request matches. Done when matching works for these cases.	P3-H1	—
P3-H3	**Enqueue the C-04 parse job** Ask the AI parser to read each new file. Done when every uploaded file is sent for parsing exactly once.	P3-H1 P3-I1	—
P3-H4	**Diagnostics and parse result API; manifest hidden from other terminals** Show the parsing result and its problems, and hide a vessel's cargo list from other terminals. Done when staff see the diagnostics and other terminals see nothing.	P3-H3	—
P3-H5	**Withdraw (409 if a confirmed plan used the file)** Let staff withdraw a file, unless a confirmed plan already used it. Done when withdrawal works and is refused in that case.	P3-H1	—
P3-H7	**Edit vessel particulars** Let staff correct the vessel's details at any time. Done when changes are saved with history.	P3-G1	—
P3-H8	**Manifest completeness gate before customs approval** Block customs approval while any line of the cargo list is incomplete. Done when an incomplete list stops the approval with a clear reason.	P3-H4	—
P3-I2	**Run row created in the status transaction; push after commit; sweeper for old QUEUED rows** Create the AI run record in the same step as the status change, send the work only after the change is saved, and re-send runs that were left waiting. Done when no run is lost or sent twice.	P3-I1 P3-G2 DEC-02 DEC-11	—
P3-I3	**Completion consumer on queue:ai-results + poller safety net; stale RUNNING shown with retry** Listen for the AI "finished" messages (CT-14) and update the screens; also check periodically for runs that finished without a message, and show runs that got stuck so staff can retry. Done when every finished run reaches the screens.	P3-I1 DEC-01	CT-14 (AI)

Backend Team · 16 tasks

ID	Task	Needs from own team	Needs from other teams
P3-I4	**SSE endpoint for the dashboard, filtered by role and terminal, within 5 s of commit** Provide the live update channel for the web dashboard, filtered by role and terminal, delivering changes within 5 seconds (CT-05). Done when a change appears on the right screens without refresh.	DEC-05	—
P3-I5	**Grants script: one database account per worker; audit_log INSERT/SELECT only** Give each AI worker its own database account with only the rights it needs, and make the audit log append-only. Done when a worker cannot touch tables it does not own.	DEC-08	—
P3-I6	**Mock workers for every queue, contract-shaped rows incl. failures, completion messages** Build stand-in AI workers that answer every queue with realistic results, including failures, in the exact contract shape (CT-19). Done when the Frontend and QA can run every AI scenario without the real AI.	P3-I1 CT-19	—
P3-I7	**Customs approval: CUSTOMS_APPROVED, part1_run QUEUED, C-01, 202** Let staff record customs approval: the request moves forward, an AI Part 1 run is created and sent, and the user gets an immediate "in progress" answer. Done when approval starts the entry-window calculation.	P3-G2 P3-I2 P3-H8	—
BE-PK4	**API readiness pack for FRR 5** Prepare the API readiness pack for the features handed off at FRR 5: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 5.	BE-PK3	—

UI/UX Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
UX-QA4	**Design QA of the screens built in Sprint 4** Compare the screens built in Sprint 4 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV4	**Revise designs from the Sprint 4 review and design QA findings** Update the designs based on the Sprint 4 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA4	—

Frontend Team · 6 tasks

ID	Task	Needs from own team	Needs from other teams
FE-047	**P-26 Port Profile and Navigation Channel** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-26 manages the port profile, the navigation channel and its waypoints. Done when the Port Admin can maintain them with validation.	FE-021 FE-032	P2-F1 (Backend) UX-W2 (UI/UX)
FE-048	**P-27 Terminal Management** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-27 manages terminals. Done when terminals can be created and edited.	FE-047	P2-F2 (Backend)
FE-049	**P-28 Berth Management (shared berths allowed)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-28 manages berths, including berths shared by several terminals, and warns that edits never change a confirmed plan. Done when berths can be maintained with every rule shown clearly.	FE-048	P2-F2 (Backend)
FE-076	**P-59 Safety Constraints** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-59 manages safety constraints and the tide data set activation. Done when the Marine Safety Officer can maintain them.	FE-021 FE-032	P2-F5 (Backend) UX-W2 (UI/UX)
FE-051	**P-62 Operational Settings** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-62 manages the port operational settings such as delay thresholds and reminder time. Done when settings can be changed without code.	FE-021 FE-032	P2-F1 (Backend) UX-W2 (UI/UX)
FE-066	**P-43 Pilot Roster and Availability (with rank limits)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-43 manages pilots, their ranks, rank limits and availability. Done when the Pilotage Manager can maintain the roster and the officer can view it.	FE-021 FE-032	P2-F3 (Backend) UX-W3 (UI/UX)

Mobile Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
MB-F4A	**Sync engine core: enqueue, ordered send, backoff ping, idempotency key reuse, DONE/REJECTED/PENDING rules, purge, counters stream** Build the engine that queues updates made offline and sends them in order, retrying safely so nothing is applied twice (DEC-41). Done when updates made offline arrive once and in order after reconnecting.	MB-F3 DEC-41	—

Mobile Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-41	**Close DEC-41: Automatic sync right after saving an event while online** Should the app sync immediately after saving an event when it is online? Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

QA Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
QA-AI3	**Collect published reference-port master data with provenance labels** Collect published data about the reference port, marking the source of every value (CT-25). Done when the data set is complete.	CT-25	—
QA-TC4	**Acceptance test cases for the features handed off at FRR 4** Write the acceptance tests for every feature handed over at FRR 4, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT4	**Sprint 4 release testing on the release branches and sign-off report** Test the Sprint 4 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC4	—

M5 · Core loop on mocks

Sprint 5 — The core loop: entry window to confirmed plan

42 tasks
Release
v0.5.0

Goal. Make the core loop work through the API with stand-in AI: customs approval, entry window, recommendation and plan confirmation, plus the tablet sync API. Part 2 starts saving real recommendations.

Handed off at the FRR

Hand-off	Approved items	To
FT-05 Tugs, equipment, yard, shifts	P2-F3 P2-F4 P2-F6 UX-W3	Frontend
FT-06 Requests and BAPLIE	P3-G1 P3-G3 P3-G4 P3-H1 P3-H4 P3-H5 P3-H7 P3-H8 P3-I7 UX-W3	Frontend
Live updates	P3-I4	Frontend

Done when

- Customs approval to confirmed plan works through the API, with conflicts refused
- Part 2 saves real recommendation versions
- Request and BAPLIE pages pass QA

Leads Council · 2 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR5	**Sprint 5 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 5, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK4 (Backend)
PL-REL5	**Sprint 5 review and release: develop → release → main, tag v0.5.0, back-merge, PortHub, demo deploy** At the end of Sprint 5, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v0.5.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT5 (QA)

AI Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
AI-056	**Solver outcome handling: FIFO fallback and NO_FEASIBLE_PLAN** Handle solver outcomes: fall back to first-come-first-served when the solver fails technically, and report "no feasible plan" with the reasons when nothing fits. Done when both paths are tested.	AI-055	—

AI Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
AI-057	**B6 template rationale and rationale factors** Write the explanation of each option from fixed templates and record the factors behind it. Done when every option has a readable, consistent explanation.	AI-055 AI-009	—
AI-058	**Recommendation persistence and version records** Save every recommendation as a version so staff can compare, override and confirm (CT-15). Done when versions are complete and traceable to their run.	AI-056 AI-057 CT-15	DEC-01 (Backend) DEC-02 (Backend)
AI-040	**A1 scheduled ingestion with provider fallback** Pull weather data on a schedule and switch to the backup provider when the main one fails. Done when data keeps flowing when one provider is down.	AI-020 AI-011	—
AI-041	**A1 on-demand pull for a Part 1 run** Pull fresh weather data on demand when a Part 1 calculation needs it. Done when each calculation uses data that is fresh enough.	AI-040	—
AI-042	**A2 astronomical tide computation (pyTMD)** Calculate the astronomical tide for the port. Done when predictions match the reference table.	AI-021 AI-011	—
AI-043	**A3 surge residual** Calculate the difference between measured sea level and the predicted tide (surge), the same way the Backend shows it. Done when both give the same number.	AI-041 AI-042	—
AI-070	**Model registry integration** Connect to the model registry so every result records which model version produced it. Done when each result is traceable to a model version.	AI-013 AI-009	—
AI-074	**Trusted-data selection (clock_skew_suspect = 0)** Select only trustworthy execution data (no suspicious device clocks) for evaluation. Done when flagged data never reaches the metrics.	AI-025	—
DEC-07	**Close DEC-07: Storing a NO_FEASIBLE_WINDOW result** blocked_period.proposal_id is mandatory, so a proposal must exist even when no window exists. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

Backend Team · 13 tasks

ID	Task	Needs from own team	Needs from other teams
P4-J1	**Part 1 result API: latest run, outcome, windows, blocked periods, sources and ages** Show the Part 1 result: the latest outcome, the safe entry windows, the blocked periods with their reasons, and where the weather data came from and how old it is. Done when every outcome (windows found, no window, no data) can be displayed.	P3-I7	DEC-07 (AI)

Backend Team · 13 tasks

ID	Task	Needs from own team	Needs from other teams
P4-J2	**PART1_RUN completion handler: SSE + ENTRY_WINDOW_READY; FAILED with recompute** When the Part 1 calculation finishes, update the request and tell the staff; if it failed, let them recompute. Done when staff are told about every outcome.	P4-J1 P3-I3	—
P4-J3	**Confirm a window (409 checks), ENTRY_WINDOW_CONFIRMED, solver_run QUEUED, C-02** Let staff confirm one entry window. The system checks it is still the latest and still valid, moves the request forward and asks the AI for a berth recommendation. Done when confirmation starts the recommendation.	P4-J1	—
P4-J4	**Reject all, recompute, reopen (409 if a plan is confirmed)** Let staff reject all windows, ask for a recalculation, or reopen the window step (not after a plan is confirmed). Done when each action behaves as the requirements state.	P4-J1	—
P4-K1	**PART2_SOLVE completion handler: RECOMMENDATION_GENERATED / NO_FEASIBLE_PLAN** When the Part 2 calculation finishes, move the request to "recommendation generated", or record that no feasible plan exists with the reasons. Done when both outcomes reach the staff.	P3-I3 DEC-02	DEC-06 (AI)
P4-K2	**Recommendation API: versions, options, tugs, rationale, FIFO flag, binding constraints; review record** Show the recommendation: its versions, the ranked options, tugs, the explanation and the constraints that shaped it, and record that the staff reviewed it. Done when staff can review every option.	P4-K1	—
P4-K3	**Regenerate before confirmation (409 after), C-02 MANUAL** Let staff ask for a new recommendation before the plan is confirmed (refused after). Done when regeneration works and the old version is replaced.	P4-K1	—
P4-L1	**Confirm plan in one transaction with lock, re-check, reservation; 409 and rollback on clash** Confirm the plan in one safe step: lock the resources, check again that they are still free, and book them. If anything clashes, nothing is saved and staff get a clear conflict message. Done when two people can never book the same resource.	P4-K2	—
P5-O1	**My tasks and GET /sync/changes (built on seeded assigned components)** Give each field user their own task list and a way for the tablet to fetch only what changed since its last sync. Done when the tablet receives exactly its own tasks.	P2-D1	—
P5-O2	**POST /sync/events: idempotent by client_event_id, ordered, rejection reason per event** Receive status updates from tablets that may have been offline: apply each update once even if it is sent twice, keep the original order, and give a clear reason for each refused update (CT-07). Done when repeated sending never duplicates an update.	P5-O1	—

Backend Team · 13 tasks

ID	Task	Needs from own team	Needs from other teams
P5-O4	**Status updates through status_type (monotonic); plan and request COMPLETED** Apply field status updates only in the allowed order, and complete the plan and the request when all their work is done. Done when statuses can never go backwards.	P5-O2	—
P5-N2	**Mobile WebSocket through Reverb, private channel per user** Provide the live update channel for the tablet, one private channel per user (CT-06). Done when a tablet receives updates meant for its user only.	P1-C3	—
BE-PK5	**API readiness pack for FRR 6** Prepare the API readiness pack for the features handed off at FRR 6: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 6.	BE-PK4	—

UI/UX Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
UX-QA5	**Design QA of the screens built in Sprint 5** Compare the screens built in Sprint 5 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV5	**Revise designs from the Sprint 5 review and design QA findings** Update the designs based on the Sprint 5 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA5	—

Frontend Team · 12 tasks

ID	Task	Needs from own team	Needs from other teams
FE-033	**SSE client and event map** Connect the app to the live update channel (CT-05) so pages refresh themselves when data changes. Done when changes appear without refresh.	FE-031	CT-05 (Backend) P3-I4 (Backend)
FE-068	**P-45 Tugboat Fleet** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-45 manages the tug fleet. Done when the Tug Manager can maintain it and the officer can view it.	FE-021 FE-032	P2-F3 (Backend) UX-W3 (UI/UX)
FE-069	**P-46 Tug Master Management (with shifts)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-46 manages tug masters, their shifts and their tug. Done when one tug master per tug is enforced on screen.	FE-068	P2-F3 (Backend)

Frontend Team · 12 tasks

ID	Task	Needs from own team	Needs from other teams
FE-071	**P-48 CT Equipment and Cranes** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-48 manages terminal equipment and cranes. Done when the CT Administrator can maintain them for their terminal.	FE-021 FE-032	P2-F4 (Backend) UX-W3 (UI/UX)
FE-072	**P-49 Container Yard Structure** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-49 manages the yard structure from zones down to slots. Done when a full yard can be entered and searched.	FE-071	P2-F4 (Backend)
FE-073	**Terminal productivity profile (tab on P-48)** Add the terminal productivity tab to P-48. Done when productivity periods can be maintained without overlaps.	FE-071	P2-F6 (Backend)
FE-078	**Port shifts settings (tab on P-62)** Add the port shifts tab to P-62. Done when shifts can be maintained.	FE-051	P2-F3 (Backend)
FE-052	**P-31 BAPLIE Files (upload, parse status, diagnostics, correction, withdraw)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-31 lets staff upload BAPLIE files, follow parsing, read diagnostics, correct and withdraw files. Done when every file status is shown clearly.	FE-032 FE-031	P3-H1 (Backend) P3-H4 (Backend) UX-W3 (UI/UX)
FE-053	**P-33 Vessel and Cargo Data (vessel edit)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-33 shows the vessel and its containers and lets staff edit vessel details. Done when staff can review the cargo and fix vessel data.	FE-052	P3-H7 (Backend)
FE-054	**P-34 Berthing Requests list** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-34 lists berthing requests with filters and live status. Done when staff can find any request quickly.	FE-031 FE-017	P3-G4 (Backend) UX-W3 (UI/UX)
FE-055	**P-35 Berthing Request Form (create and edit)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-35 creates and edits berthing requests. Done when a complete request can be submitted and errors are clear.	FE-016 FE-032	P3-G1 (Backend) UX-W3 (UI/UX)
FE-056	**P-36 Request Details: review decisions and customs approval** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-36 shows one request with its timeline and lets staff take review decisions and record customs approval. Done when every decision available in each status works.	FE-055 FE-017	P3-G3 (Backend) P3-I7 (Backend)

Mobile Team · 1 tasks

ID	Task	Needs from own team	Needs from other teams
MB-P56	**P-56 Sync Status and Outbox screen** Build P-56, which shows waiting, sent and refused updates with their reasons. Done when the user always knows what has not reached the office yet.	MB-F4A	UX-W1 (UI/UX)

QA Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
QA-TC5	**Acceptance test cases for the features handed off at FRR 5** Write the acceptance tests for every feature handed over at FRR 5, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT5	**Sprint 5 release testing on the release branches and sign-off report** Test the Sprint 5 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC5	—

M6 · Real AI in the loop · Surge sprint

Sprint 6 — Dispatch, field execution and real AI

63 tasks
Release v0.6.0

Goal. The surge sprint carries more work than the others. The real BAPLIE parser and Part 2 join the loop, Part 1 is completed, and the Backend builds dispatch, notifications, deviations and yard. The Frontend builds the AI review pages and the Mobile Team builds sync and the task list.

Handed off at the FRR

Hand-off	Approved items	To
FT-07 Entry window (Part 1)	P4-J1 P4-J3 P4-J4 UX-W4	Frontend
FT-08 Recommendation and plan (Part 2)	P4-K1 P4-K2 P4-L1 UX-W4	Frontend
FT-09 Tablet sync and tasks	P5-O1 P5-O2 P5-N2 UX-W4	Mobile

Done when

- The real BAPLIE parser and Part 2 pass the contract tests
- Part 1 produces all three outcomes
- P-37, P-38, P-39 and My Tasks pass QA
- Dispatch, notification, deviation and yard APIs are merged

Leads Council · 2 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR6	**Sprint 6 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 6, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK5 (Backend)
PL-REL6	**Sprint 6 review and release: develop → release → main, tag v0.6.0, back-merge, PortHub, demo deploy** At the end of Sprint 6, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v0.6.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT6 (QA)

AI Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
AI-059	**B7 re-planning: resource change, regenerate, delay re-plan** Re-plan when a resource changes, when staff ask for a new recommendation, and when a delay happens, keeping work already started fixed. Done when each trigger produces a valid new version.	AI-058	—
AI-060	**Override and adjustment validation service (per OD-03)** Check a staff override or yard adjustment against the hard rules and explain any rule it breaks (DEC-03, DEC-18). Done when invalid changes are always refused.	AI-051 DEC-03 DEC-18	—

AI Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
AI-045	**A5 UKC with Barrass squat (domain reviewer sign-off)** Calculate the water under the keel, including the squat effect, and have it reviewed by a domain expert. Done when results match reviewed hand calculations.	AI-043 AI-009	—
AI-047	**A7 demand aggregation and peak congestion score** Count expected arrivals per time slot and give a congestion score. Done when scores match the open requests.	AI-013 DEC-24	—
AI-048	**Part 1 run orchestrator and persistence of outcomes** Run the full Part 1 calculation for a request and save its outcome (windows found, no window, or no data) (DEC-07, DEC-14, DEC-24). Done when all three outcomes are produced on scripted scenarios.	AI-041 AI-044 AI-045 AI-047 DEC-07 DEC-24 CT-08 DEC-14	—
AI-066	**Yard placement generator and persistence** Generate the yard placement for a container vessel and save it (DEC-16). Done when placements respect every yard rule.	AI-065 AI-032 AI-058 DEC-16	—
AI-071	**C1 outcome evaluation on trusted data** Measure how good the AI results were against what really happened, on trusted data only. Done when metrics are stored for each period.	AI-070 AI-074	—
AI-072	**C2 drift monitor and drift alert record** Watch for drift in the AI inputs and results and raise an alert above the agreed limit. Done when a drift test raises the alert.	AI-070 AI-008	—
AI-073	**Part 3 scheduling inside the Part 3 worker** Run the Part 3 evaluation on its own schedule inside its worker (DEC-31). Done when it runs without anyone starting it.	AI-011 DEC-31	—
AI-080	**AI/Backend contract tests (both CIs)** Run automatic contract checks in both the AI and Backend pipelines so neither side can break the agreed messages (CT-08). Done when a contract change fails the build on both sides.	CT-08 AI-048 AI-058 AI-032	—
AI-092	**Observability: structured logs with run id and metrics** Make every worker log in a structured way with the run number, and expose basic counts. Done when any run can be followed through the logs.	AI-011	—
AI-022	**Reference-port master-data dataset for testing (assembled with the QA Team)** Assemble the reference port master data for testing from published sources, with the QA Team, marking anything generated (CT-25). Done when the data set loads and every row shows its source.	AI-015	QA-AI3 (QA)
DEC-31	**Close DEC-31: Part 3 scheduling host** There is no queue:part3. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

AI Team · 15 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-43	**Close DEC-43: Recommendation confidence field on P-38** There is no stored confidence field for a recommendation. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
AI-F03	**FastAPI: validate endpoint (override and yard adjustment)** Expose the instant check for overrides and yard adjustments on the internal service (CT-13). Done when the Backend gets an answer with the broken rule, or a clear refusal when the service is down.	AI-F01 AI-060 CT-13	—

Backend Team · 31 tasks

ID	Task	Needs from own team	Needs from other teams
P4-K4	**Override one component with reason; MODIFIED_BY_STAFF version (per D-03)** Let staff override one part of a recommendation with a reason. The change is checked against the hard rules before it is accepted (DEC-03), and saved as a new version. Done when an invalid override is refused with the broken rule.	P4-K2	DEC-03 (AI) CT-13 (AI)
P4-K5	**Resource change listener: regenerate pending recommendations (C-02 RESOURCE_CHANGE)** When a resource changes, ask the AI to refresh only the pending recommendations that use it. Done when confirmed plans are never touched and affected pending ones are refreshed.	P4-K1 P2-F7	—
P4-J5	**Manual environmental data: completeness and horizon check, MANUAL forecast run, C-01** Let staff enter weather data by hand when the providers fail. The data must be complete and cover the needed period, and it goes through the same safety rules. Done when manual data can produce entry windows.	P4-J1	—
P4-J6	**Current conditions view with surge, source, age and stale warning** Show the current conditions with the tide surge, the data source, its age and a warning when the data is too old. Done when stale data is clearly flagged.	P4-J1	—
P4-L2	**Cancel before any start: release resources, PLAN_CANCELLED, continue → new C-02** Let staff cancel a plan before any work started: release the resources, mark the plan cancelled, and if the vessel still comes, ask for a new recommendation. Done when cancellation frees everything it held.	P4-L1	—
P4-L3	**Port Admin safety exception after a start (SAFETY_EXCEPTION audit)** Let the Port Admin stop the remaining work of a started plan for safety reasons, with a mandatory reason that is audited. Done when only the Port Admin can do it and it is fully recorded.	P4-L2	—
P4-L4	**Unified port state view: berths, tugs, pilots, requests, plans** Give one live view of the whole port: berths, tugs, pilots, requests and plans. Done when the view matches the real state at any time.	P4-L1	—

Backend Team · 31 tasks

ID	Task	Needs from own team	Needs from other teams
P5-M1	**Propose pilot, tug master and tug from shifts, schedules and reservations** Propose a pilot, tug master and tugs for each confirmed plan part, based on shifts, schedules and existing bookings. Done when every plan part gets a sensible proposal.	P4-L1	—
P5-M2	**Officer confirms or replaces: availability, rank vs LOA, shift, conflicts (422/409)** Let the officer confirm or replace the proposal. A replacement is checked for availability, rank against vessel size, shift and clashes, with clear reasons when refused (DEC-37). Done when invalid choices are refused.	P5-M1 DEC-37	—
P5-M3	**Save assignment with audit; component ASSIGNED** Save the assignment with its audit and mark the plan part as assigned. Done when assignments appear in the plan and in the field user's tasks.	P5-M2	—
P5-N1	**Notification service: after commit, notification_log per send, WEB and MOBILE, 12 types** Send notifications after every relevant change, to the web and the tablet, and log every send. Done when all agreed notification types are delivered and logged.	P3-I4	—
P5-N3	**Re-send T+10 / T+20, ACK_ESCALATION at T+30; cancelled on acknowledgement** Remind field users who did not acknowledge a task (two reminders) and then escalate to the officer and manager. Acknowledging stops the reminders. Done when an unacknowledged task is always escalated.	P5-N1	—
P5-N4	**Pre-execution reminder at the port_setting lead time (default 2 h)** Remind field users before their task starts, at a lead time the Port Admin can change. Done when reminders arrive on time.	P5-N1	—
P5-O3	**Acknowledgement: notification ACKNOWLEDGED, retries cancelled, officer sees it by SSE** Record when a field user acknowledges a task, stop the reminders, and show it to the officer live. Done when the officer sees acknowledgements immediately.	P5-O2 P5-N3 DEC-40	—
P5-O5	**Store client_sent_at; flag clock_skew_suspect above 5 minutes** Keep the time an update was sent from the device and flag updates whose device clock looks wrong. Done when suspicious updates are flagged and excluded from AI evaluation.	P5-O4 DEC-44	—
P5-O6	**Deviation detection into deviation_log; thresholds as port_setting keys; browse and filter** Detect differences between planned and actual times, using thresholds the Port Admin sets (DEC-09), and let staff browse them. Done when every significant difference is recorded.	P5-O4 DEC-09	—
P5-O7	**Significant delay: C-02 REPLAN; confirm or discard proposal in one transaction** When a significant delay happens, ask the AI for a new plan proposal; staff confirm or discard it in one safe step. Done when a confirmed proposal replaces the plan without breaking work already started.	P5-O6	—

Backend Team · 31 tasks

ID	Task	Needs from own team	Needs from other teams
P5-O8	**Terminal supervisor status from the tablet, same deviation path** Let the terminal supervisor report terminal status from the tablet, feeding the same delay detection (DEC-39). Done when supervisor updates create deviations like other field updates.	P5-O4 DEC-39	—
P6-P1	**On plan confirm (container vessel, ACTIVE BAPLIE): YARD_REC_REQUESTED and C-03** When a plan is confirmed for a container vessel with an active cargo file, ask the AI for a yard placement automatically. Done when every such plan gets a yard request.	P4-L1	—
P6-P2	**Yard result API for the owning terminal; YARD_RECOMMENDATION_READY / _FAILED** Show the yard placement result to its own terminal only, and tell the planner when it is ready or failed. Done when other terminals cannot see it.	P6-P1	—
P6-P3	**Accept (no reservation), adjust (re-validated), reject** Let the planner accept (no reservation), adjust (each change re-checked) or reject the yard placement. Done when invalid adjustments are refused with the broken rule.	P6-P2	CT-13 (AI)
P6-P4	**TOS-ready view; record EXPORTED_TO_TOS** Give the planner a version of the accepted placement that can be typed into the terminal system, and record that it was exported. Done when the export is recorded.	P6-P3	—
P3-H6	**Correct staged BAPLIE data and re-validate (per D-03)** Let staff correct a file that was held for problems and check it again with the same rules the parser uses (DEC-03). Done when a corrected file can become active.	P3-H4	DEC-03 (AI)
P3-I8	**Contract tests in Backend CI against the contract v1 JSON Schemas** Run automatic checks that every message the Backend sends matches the agreed AI contracts. Done when a contract break fails the build.	P3-I1	CT-08 (AI)
P6-R6	**Prototype server: model volume, BAPLIE volume, supervisor for workers and consumer** Prepare the prototype server: storage for models and cargo files, and automatic restart for the AI workers and the result listener. Done when the server survives a restart without manual work.	P3-I3	—
DEC-09	**Close DEC-09: Where deviation thresholds live** The thresholds that make a delay "significant" per operation type have no place in the schema. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-37	**Close DEC-37: "Reject with reason" on P-44 and P-47** The Backend tasks only have confirm or replace. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—

Backend Team · 31 tasks

ID	Task	Needs from own team	Needs from other teams
DEC-39	**Close DEC-39: CT Supervisor status list** P-55 says the terminal status list needs confirmation. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-40	**Close DEC-40: Mobile Acknowledge event type and payload** Acknowledge goes through the Outbox and needs an agreed event type. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
DEC-44	**Close DEC-44: Storing the clock-skew review note** P-41 shows a Clock-skew suspect flag; where a review note is stored is unclear. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
BE-PK6	**API readiness pack for FRR 7** Prepare the API readiness pack for the features handed off at FRR 7: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 7.	BE-PK5	—

UI/UX Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
UX-QA6	**Design QA of the screens built in Sprint 6** Compare the screens built in Sprint 6 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV6	**Revise designs from the Sprint 6 review and design QA findings** Update the designs based on the Sprint 6 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA6	—

Frontend Team · 8 tasks

ID	Task	Needs from own team	Needs from other teams
FE-057	**P-37 Entry-Time Window Review (AI Part 1)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-37 shows the AI entry windows, blocked periods and data freshness, and lets staff confirm a window, reject all or recompute (AI Part 1). Done when every Part 1 outcome is shown and handled.	FE-056	P4-J1 (Backend) P4-J3 (Backend) P4-J4 (Backend) UX-W4 (UI/UX) CT-20 (Backend)

Frontend Team · 8 tasks

ID	Task	Needs from own team	Needs from other teams
FE-059	**P-38 Recommendation Review: options and rationale (AI Part 2)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-38 shows the AI recommendation options with their explanation and lets staff choose one (AI Part 2). Done when staff can compare options and see every recommendation state.	FE-057 FE-017	P4-K2 (Backend) UX-W4 (UI/UX) DEC-43 (AI) CT-20 (Backend)
FE-061	**Confirm the plan (human gate 2)** Let staff confirm the chosen option as the plan, and handle the case where a resource was taken meanwhile. Done when confirmation and its conflict case both work.	FE-059	P4-L1 (Backend)
FE-062	**P-39 Operational Plans list** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-39 lists operational plans with live status; terminal users see only their terminal, read-only. Done when every role sees the right plans.	FE-031 FE-017	P4-L1 (Backend) UX-W4 (UI/UX)
FE-101	**API contract tests** Test automatically that the app and the API still agree (DEC-47). Done when an API change that breaks the app fails the pipeline.	FE-035 FE-034	CT-02 (Backend) DEC-47 (Backend)
FE-102	**Role and access matrix tests** Test that every role sees and does only what it is allowed. Done when the role matrix passes.	FE-021 FE-022	—
FE-103	**Error-state tests** Test every error state: not signed in, not allowed, conflict, invalid input, old app version. Done when each shows the right message.	FE-030 FE-032 FE-033 FE-034	—
FE-115	**Port web build, NGINX hand-off and signed build** Produce the production build and hand it to the web server setup. Done when the demo server runs the build.	FE-011 FE-012	—

Mobile Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
MB-F4B	**Sync engine integration: POST /sync/events, GET /sync/changes, 401/426 stop, same-user rule** Connect the sync engine to the server: send queued updates and fetch what changed, and stop safely when the session expires or the app is outdated (CT-07, DEC-47). Done when sync works with the real server.	MB-F4A	P5-O1 (Backend) P5-O2 (Backend) DEC-47 (Backend) CT-07 (Backend)
MB-P54	**P-54 My Tasks: cache-first list, task cards, plan-changed notice, role content** Build P-54, the field user's task list, working from the cache when offline and updating live. Done when each role sees its own tasks online and offline.	MB-F4B	P5-O1 (Backend) UX-W4 (UI/UX)

Mobile Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
MB-F5	**Realtime: Reverb WebSocket, event id then REST fetch, de-duplication, catch-up, Android foreground service** Receive live updates (CT-06), catch up after reconnecting, and keep updates flowing while a task is active. Done when changes reach the tablet without refresh.	MB-F2	P5-N2 (Backend) CT-06 (Backend)

QA Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
QA-TC6	**Acceptance test cases for the features handed off at FRR 6** Write the acceptance tests for every feature handed over at FRR 6, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT6	**Sprint 6 release testing on the release branches and sign-off report** Test the Sprint 6 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC6	—

M7 · Feature-complete MVP

Sprint 7 — Yard, monitoring and end-to-end integration

35 tasks
Release
v0.7.0

Goal. Integrate everything: replace every stand-in with the real AI, run the eight main workflows end to end, and finish the remaining web and tablet pages.

Handed off at the FRR

Hand-off	Approved items	To
FT-08 Override, cancel, safety exception	P4-K3 P4-K4 P4-L2 P4-L3 P4-J5 UX-W4	Frontend
FT-09 Dispatch, notifications, status	P5-M1 P5-M2 P5-M3 P5-N1 P5-N3 P5-O3 P5-O4 UX-W5	Frontend · Mobile
FT-10 Deviations, re-plan, port state	P5-O6 P5-O7 P4-L4 P4-J6 UX-W5	Frontend
FT-11 Yard and TOS hand-off	P6-P1 P6-P2 P6-P3 P6-P4 UX-W6	Frontend

Done when

- No stand-in worker is left in the integrated build
- WF-1 to WF-8 pass on the demo environment
- Offline tablet updates arrive once
- Every MVP page is built

Leads Council · 2 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR7	**Sprint 7 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 7, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK6 (Backend)
PL-REL7	**Sprint 7 review and release: develop → release → main, tag v0.7.0, back-merge, PortHub, demo deploy** At the end of Sprint 7, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v0.7.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT7 (QA)

AI Team · 8 tasks

ID	Task	Needs from own team	Needs from other teams
AI-049	**Part 1 validation suite (golden scenarios)** Build a set of known Part 1 scenarios with expected results. Done when the suite runs in the pipeline.	AI-048 AI-022	—

AI Team · 8 tasks

ID	Task	Needs from own team	Needs from other teams
AI-062	**Part 2 validation suite and baseline-comparison harness** Build a set of known Part 2 scenarios and compare results with the simple baseline. Done when the suite runs and reports the comparison.	AI-059 AI-015	—
AI-067	**Yard adjustment re-validation** Re-check every yard adjustment against the yard rules. Done when invalid adjustments are refused.	AI-066 DEC-03	—
AI-068	**Yard validation suite** Build a set of known yard scenarios with expected results. Done when the suite runs in the pipeline.	AI-066	—
AI-081	**End-to-end workflow tests WF-1 to WF-8 with the Backend** Run the eight main workflows from start to finish with the Backend. Done when all eight pass on the demo environment.	AI-080 AI-059 AI-066 AI-071 AI-072	—
AI-090	**Package and deploy the four workers on the prototype server** Package and deploy the four workers on the prototype server with automatic restart. Done when all workers run there after a reboot.	AI-081	—
DEC-33	**Close DEC-33: Manual-planning baseline** Evaluating the AI needs a comparison with manual planning. Agree on one of the options in the Decisions file, record the decision and tell every affected team. Done when the decision is written in the Decisions file before the tasks that depend on it start.	—	—
AI-F04	**Deploy the FastAPI service on the prototype server with the workers** Deploy the internal AI service next to the workers on the prototype server. Done when it runs there and restarts automatically.	AI-F03 AI-F02	—

Backend Team · 6 tasks

ID	Task	Needs from own team	Needs from other teams
P6-R1	**Swap the mock for each delivered worker (EDI, Part 2, Part 1); contract tests green** Replace each stand-in worker with the real AI worker as it becomes ready, with the contract checks passing on both sides. Done when no stand-in is left in the integrated build.	P3-I6	AI-080 (AI)
P6-R2	**Evaluation and drift data for the Port Admin; DRIFT_ALERT from Part 3** Provide the AI evaluation and drift data to the Port Admin and raise an alert when the AI reports drift. Done when the governance page has real data.	P3-I3	—
P6-R3	**End-to-end workflows WF-1 to WF-8 with the AI Team; SSE delay measured** Run all eight main workflows from start to finish together with the AI Team, and measure how fast live updates arrive. Done when all eight pass on the demo environment.	P6-R1	—

Backend Team · 6 tasks

ID	Task	Needs from own team	Needs from other teams
P6-R4	**Failure paths: worker down, stale run, solver timeout to FIFO, 409 race, Redis down** Test what happens when things fail: a worker stops, a run gets stuck, the solver runs out of time, two people book the same resource, Redis goes down. Done when each case ends in the documented safe state.	P6-R1	—
BE-SUP7	**Integration support for Frontend and Mobile defects found in Sprint 7** Keep time free to fix problems the Frontend and Mobile teams find while connecting to the real APIs. Done when no integration defect is left open at the sprint review.	—	—
BE-PK7	**API readiness pack for FRR 8** Prepare the API readiness pack for the features handed off at FRR 8: the endpoints, examples, demo data, live events and known gaps (CT-23). Done when the pack is ready before FRR 8.	BE-PK6	—

UI/UX Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
UX-QA7	**Design QA of the screens built in Sprint 7** Compare the screens built in Sprint 7 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV7	**Revise designs from the Sprint 7 review and design QA findings** Update the designs based on the Sprint 7 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA7	—

Frontend Team · 12 tasks

ID	Task	Needs from own team	Needs from other teams
FE-063	**P-40 Operational Plan Details: cancel, safety exception, versions** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-40 shows one plan with its parts and progress, and supports cancellation, the Port Admin safety exception and version history. Done when every plan action works.	FE-062	P4-L2 (Backend) P4-L3 (Backend)
FE-058	**Manual environmental data entry (AI Part 1)** Add manual weather data entry to the entry-window flow for when providers fail. Done when manual data produces new windows.	FE-057	P4-J5 (Backend)
FE-060	**Override one component and regenerate (AI Part 2)** Let staff override one part of a recommendation with a reason and ask for a new recommendation. Done when invalid overrides show the broken rule.	FE-059	P4-K3 (Backend) P4-K4 (Backend)

Frontend Team · 12 tasks

ID	Task	Needs from own team	Needs from other teams
FE-067	**P-44 Pilotage Assignments (confirm or replace proposal)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-44 lets the pilotage officer confirm or replace the proposed pilot. Done when invalid choices show the reason.	FE-066 FE-033	P5-M2 (Backend) UX-W5 (UI/UX)
FE-070	**P-47 Tug Assignments (confirm or replace proposal)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-47 lets the tug officer confirm or replace the proposed tug master and tugs. Done when invalid choices show the reason.	FE-069 FE-033	P5-M2 (Backend) UX-W5 (UI/UX)
FE-042	**P-19 Notifications (web)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-19 lists notifications and lets users acknowledge them and open the related item. Done when new notifications appear live.	FE-020 FE-033	P5-N1 (Backend)
FE-065	**P-41 Deviation Records and P-42 Deviation Details** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-41 lists differences between planned and actual times; P-42 shows one in detail with its trust flag. Done when staff can review every deviation.	FE-031 FE-017	P5-O6 (Backend) UX-W5 (UI/UX)
FE-064	**Delay re-plan proposal on P-40 (AI Part 2)** Show the AI delay re-plan proposal on P-40 and let staff confirm or discard it. Done when a confirmed proposal updates the plan.	FE-063	P5-O7 (Backend)
FE-041	**P-18 Unified Port State** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-18 shows the live state of the whole port. Done when it updates live and respects terminal scope.	FE-020 FE-033	P4-L4 (Backend) UX-W5 (UI/UX)
FE-043	**P-21 Environmental Conditions** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-21 shows weather, tide and data freshness with stale warnings. Done when old data is clearly flagged.	FE-020 FE-031	P4-J6 (Backend) UX-W5 (UI/UX)
FE-074	**P-50 Yard Placement Recommendations (AI Part 2)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-50 lists yard placement recommendations for the planner's terminal. Done when the planner sees every recommendation state.	FE-031 FE-017 FE-033	P6-P2 (Backend) UX-W6 (UI/UX) CT-20 (Backend)
FE-075	**P-51 Yard Placement Review and TOS hand-off** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-51 lets the planner accept, adjust or reject a yard placement and record the export to the terminal system. Done when invalid adjustments show the broken rule.	FE-074	P6-P3 (Backend) P6-P4 (Backend)

Mobile Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
MB-P55	**P-55 Task Details and Status Update: Acknowledge, In progress / Executed, offline save, 409, rejected events** Build P-55, where the field user acknowledges a task and records In progress and Executed in a few taps, saved on the device first (DEC-39, DEC-40). Done when updates work offline and refused updates show their reason.	MB-P54	P5-O4 (Backend) P5-O3 (Backend) UX-W4 (UI/UX) DEC-39 (Backend) DEC-40 (Backend)
MB-P19	**P-19 Notifications (tablet): list, acknowledge via Outbox, open related task** Build the tablet version of P-19: notifications, acknowledgement through the update queue, and opening the related task. Done when notifications arrive live and work offline.	MB-F5	P5-N1 (Backend) UX-W4 (UI/UX)

QA Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
QA-TC7	**Acceptance test cases for the features handed off at FRR 7** Write the acceptance tests for every feature handed over at FRR 7, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT7	**Sprint 7 release testing on the release branches and sign-off report** Test the Sprint 7 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC7	—
QA-E2E	**Cross-team end-to-end suite WF-1 to WF-8 on the demo environment** Run the eight main workflows from start to finish across all teams on the demo environment. Done when all pass or defects are logged.	—	—

M8 · Release v1.0.0

Sprint 8 — Hardening, regression and final release

33 tasks
Release v1.0.0

Goal. Harden and release: failure, performance, security and regression tests, the Part 3 governance page, field tests, the demo app and the final v1.0.0 release with the handover package.

Handed off at the FRR

Hand-off	Approved items	To
FT-12 Evaluation and drift (Part 3)	P6-R2 UX-W6	Frontend
Release candidate	The integrated Sprint 7 build with no stand-in workers.	All teams

Done when

- Failure, performance and regression tests are run and reported
- Known limitations are documented
- v1.0.0 is tagged on main in all four repositories; PortHub and demo are updated
- The demo is rehearsed with the acceptance script

Leads Council · 3 tasks

ID	Task	Needs from own team	Needs from other teams
PL-FRR8	**Sprint 8 Feature Readiness Review: approve APIs and UI and hand them to Frontend and Mobile** At the start of Sprint 8, the team leads check the API readiness pack and the final screens for each feature planned for this sprint, approve them, and hand them to the Frontend and Mobile teams. Features that are not ready stay with their owners. Done when each hand-off is recorded in the tracker.	—	BE-PK7 (Backend)
PL-REL8	**Sprint 8 review and release: develop → release → main, tag v1.0.0, back-merge, PortHub, demo deploy** At the end of Sprint 8, review the sprint with all teams and publish the release: move the tested work from develop to release to main, tag it v1.0.0 in every repository, bring fixes back to develop, update PortHub and the demo. Done when the tag exists and the demo runs it.	—	QA-RT8 (QA)
PL-06	**Final demo, v1.0.0 tag and handover pack** Run the final demo, tag version 1.0.0 and hand over the documentation package. Done when the release is tagged and the handover is complete.	PL-REL8	—

AI Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
AI-033	**Parser performance against NFR-PERF-2** Measure how fast files of each size class are parsed against the targets. Done when results are recorded and targets are met or gaps are listed.	AI-032 AI-008	—
AI-061	**Part 2 latency and concurrency measurement** Measure Part 2 speed and behaviour with several requests at once. Done when results are recorded against the targets.	AI-059 AI-015	—

AI Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
AI-082	**Failure-path and recovery tests** Test recovery from failures: provider outage, old data, solver timeout, worker crash, duplicate jobs. Done when each case ends in the documented safe state.	AI-081	—
AI-083	**Security and isolation tests** Test security and isolation: database rights, no unexpected outgoing traffic, no terminal data leaking into explanations. Done when no violation is found.	AI-081	—
AI-084	**Evaluation against FIFO and manual-planning baselines** Compare the AI with first-come-first-served and with the manual planning baseline on agreed scenarios, and report the results as measured (DEC-33). Done when the report is published.	AI-062 AI-008 DEC-33	—
AI-091	**Worker health checks and heartbeats** Give each worker a health check and a heartbeat so a stopped worker is noticed. Done when a stopped worker shows as down.	AI-090	—
AI-093	**Reproducibility: pinned dependencies and run replay** Pin library versions and make it possible to replay any run from its stored inputs. Done when a replay gives the same result.	AI-058 AI-048	—
AI-094	**Define and rehearse rollback** Write the rollback steps for the AI and rehearse them once. Done when the rehearsal succeeds.	AI-090 AI-070	—
AI-095	**AI developer and operations documentation** Write the developer and operations documentation for the AI. Done when a new member can run and operate the workers from it.	AI-090	—
AI-024	**Assess historical operational data and data-sufficiency criteria** Assess what historical operational data exists and when there will be enough to train models later. Done when the assessment is written.	AI-008	—

Backend Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
P6-R5	**Performance: API p95, SSE within 5 s, solver path within 12 s** Measure performance: API response times, live update delay and the full recommendation path against the targets. Done when results are recorded and targets are met or gaps are listed.	P6-R3	—
BE-SUP8	**Bug fixing, hardening and release support** Fix the remaining defects and harden the Backend for the final release. Done when the release test passes with no blocking defect.	—	—

UI/UX Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
UX-08	**Final polish, demo storyboard and future-work screens archive (SaaS, licence, NCT)** Polish the final screens, prepare the demo storyboard and archive the future-work screens. Done when the demo storyboard is approved.	—	—
UX-QA8	**Design QA of the screens built in Sprint 8** Compare the screens built in Sprint 8 with the approved designs and list every difference. Done when the findings are shared with the Frontend and Mobile teams.	—	—
UX-RV8	**Revise designs from the Sprint 8 review and design QA findings** Update the designs based on the Sprint 8 review and the design QA findings, and share the changes with the Frontend and Mobile teams. Done when every agreed change is in the design file.	UX-QA8	—

Frontend Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
FE-077	**P-30 AI Model Governance: evaluation and drift (AI Part 3)** Build the page as approved in the Screens Specification and the final UI, with its loading, empty and error states. P-30 shows the AI evaluation and drift for the Port Admin (AI Part 3). Done when real evaluation data is displayed.	FE-021 FE-031	P6-R2 (Backend) UX-W6 (UI/UX) CT-20 (Backend)
FE-105	**End-to-end core loop** Test the core flow end to end: request, approvals, entry window, recommendation, plan and assignments. Done when the flow passes on the demo environment.	FE-056 FE-057 FE-061 FE-063 FE-067 FE-070 FE-101	—
FE-107	**Accessibility verification** Check the agreed accessibility level on the main pages. Done when the checklist passes or gaps are listed.	FE-016 FE-006	—
FE-108	**Responsive layout verification** Check the layout on the agreed screen sizes. Done when no page breaks on them.	FE-006 FE-020	—
FE-109	**Realtime latency test (NFR-PERF-4)** Measure how fast live updates reach the screen against the target. Done when the result is recorded.	FE-033	CT-05 (Backend)
FE-110	**Terminal isolation tests** Test that terminal users never see another terminal's data. Done when isolation tests pass.	FE-072 FE-075 FE-062	—
FE-111	**Frontend security review** Review the web app for common security problems. Done when findings are fixed or listed.	FE-025 FE-045	—

Frontend Team · 10 tasks

ID	Task	Needs from own team	Needs from other teams
FE-112	**Frontend performance budget** Set and check limits for page size and load time. Done when the main pages meet the limits.	FE-115	—
FE-113	**Regression suite and release checklist** Collect the regression tests and the release checklist for the final release. Done when the checklist is complete.	FE-101 FE-102 FE-103 FE-105 FE-107 FE-110 FE-111	—
FE-118	**Release verification and known limitations** Verify the final release and write the list of known limitations. Done when the release is signed off.	FE-113 FE-115 FE-109	—

Mobile Team · 2 tasks

ID	Task	Needs from own team	Needs from other teams
MB-R1	**Field test suite: airplane mode, duplicate send, cancelled plan offline, token expiry, 426, logout, each role, phone and tablet sizes** Test the app in the field: airplane mode, double sending, a plan cancelled while offline, expired session, outdated app, each role and phone and tablet sizes. Done when every scenario behaves as specified.	MB-P55 MB-P19	—
MB-R2	**Demo environment with the Backend Team and demo APK build** Prepare the demo environment with the Backend Team and build the demo app. Done when the demo app installs and connects to the demo server.	MB-R1	—

QA Team · 3 tasks

ID	Task	Needs from own team	Needs from other teams
QA-TC8	**Acceptance test cases for the features handed off at FRR 8** Write the acceptance tests for every feature handed over at FRR 8, based on the requirements and the screens. Done when every handed-over feature has its tests.	—	—
QA-RT8	**Sprint 8 release testing on the release branches and sign-off report** Test the Sprint 8 release branch with the acceptance and regression tests and write the sign-off report. Done when the report is published and no blocking defect is open.	QA-TC8	—
QA-UAT	**Final regression, UAT script and demo rehearsal** Run the final regression, write the user acceptance script and rehearse the demo. Done when the rehearsal passes.	—	—

SECTION 3

All cross-team dependencies

Every case where a task needs something from another team. "Same sprint" means the hand-over must happen early in the sprint so the receiving team has time to work.

Task	Team	Sprint	Needs	From	Sprint	Gap
CT-14	AI	1	DEC-01	Backend	1	Same sprint
CT-15	AI	1	DEC-02	Backend	1	Same sprint
CT-19	Backend	1	CT-08	AI	1	Same sprint
DEC-19	AI	1	DEC-01	Backend	1	Same sprint
FE-015	Frontend	1	UX-W1	UI/UX	1	Same sprint
MB-P20	Mobile	1	UX-W1	UI/UX	1	Same sprint
MB-P57	Mobile	1	UX-W1	UI/UX	1	Same sprint
AI-008	AI	2	QA-AI1	QA	1	1 sprint earlier
AI-011	AI	2	DEC-01	Backend	1	1 sprint earlier
AI-013	AI	2	DEC-08	Backend	1	1 sprint earlier
AI-013	AI	2	CT-17	Backend	1	1 sprint earlier
DEC-30	AI	2	QA-AI1	QA	1	1 sprint earlier
FE-016	Frontend	2	CT-22	UI/UX	1	1 sprint earlier
FE-017	Frontend	2	CT-21	Backend	1	1 sprint earlier
FE-021	Frontend	2	DEC-48	Leads	2	Same sprint
FE-025	Frontend	2	P1-C1	Backend	1	1 sprint earlier
FE-025	Frontend	2	UX-W1	UI/UX	1	1 sprint earlier
FE-025	Frontend	2	CT-03	Backend	1	1 sprint earlier
FE-026	Frontend	2	P1-C2	Backend	1	1 sprint earlier
FE-030	Frontend	2	CT-01	Backend	1	1 sprint earlier
MB-F6	Mobile	2	UX-W1	UI/UX	1	1 sprint earlier
MB-F6	Mobile	2	CT-22	UI/UX	1	1 sprint earlier
P1-B6	Backend	2	CT-18	AI	1	1 sprint earlier
PL-FRR2	Leads	2	BE-PK1	Backend	1	1 sprint earlier
PL-FRR2	Leads	2	CT-22	UI/UX	1	1 sprint earlier
PL-REL2	Leads	2	QA-RT2	QA	2	Same sprint
AI-023	AI	3	QA-AI2	QA	2	1 sprint earlier
FE-035	Frontend	3	CT-02	Backend	1	2 sprints earlier

Task	Team	Sprint	Needs	From	Sprint	Gap
FE-045	Frontend	3	P2-E1	Backend	2	1 sprint earlier
FE-045	Frontend	3	P2-E2	Backend	2	1 sprint earlier
FE-045	Frontend	3	UX-W2	UI/UX	1	2 sprints earlier
FE-046	Frontend	3	P2-E3	Backend	2	1 sprint earlier
FE-050	Frontend	3	P2-F8	Backend	2	1 sprint earlier
FE-050	Frontend	3	UX-W2	UI/UX	1	2 sprints earlier
MB-P52	Mobile	3	P1-C3	Backend	2	1 sprint earlier
MB-P52	Mobile	3	CT-04	Backend	1	2 sprints earlier
MB-P53	Mobile	3	P1-C5	Backend	2	1 sprint earlier
MB-P53	Mobile	3	CT-04	Backend	1	2 sprints earlier
MB-P58	Mobile	3	P1-C4	Backend	2	1 sprint earlier
P2-F5	Backend	3	DEC-38	UI/UX	3	Same sprint
P2-F9	Backend	3	AI-009	AI	2	1 sprint earlier
P2-F9	Backend	3	CT-18	AI	1	2 sprints earlier
P3-I1	Backend	3	CT-08	AI	1	2 sprints earlier
P3-I1	Backend	3	DEC-04	AI	1	2 sprints earlier
P3-I1	Backend	3	CT-09	AI	1	2 sprints earlier
P3-I1	Backend	3	CT-10	AI	1	2 sprints earlier
P3-I1	Backend	3	CT-11	AI	1	2 sprints earlier
P3-I1	Backend	3	CT-12	AI	1	2 sprints earlier
PL-FRR3	Leads	3	BE-PK2	Backend	2	1 sprint earlier
PL-REL3	Leads	3	QA-RT3	QA	3	Same sprint
FE-047	Frontend	4	P2-F1	Backend	3	1 sprint earlier
FE-047	Frontend	4	UX-W2	UI/UX	1	3 sprints earlier
FE-048	Frontend	4	P2-F2	Backend	3	1 sprint earlier
FE-049	Frontend	4	P2-F2	Backend	3	1 sprint earlier
FE-051	Frontend	4	P2-F1	Backend	3	1 sprint earlier
FE-051	Frontend	4	UX-W2	UI/UX	1	3 sprints earlier
FE-066	Frontend	4	P2-F3	Backend	3	1 sprint earlier
FE-066	Frontend	4	UX-W3	UI/UX	1	3 sprints earlier
FE-076	Frontend	4	P2-F5	Backend	3	1 sprint earlier
FE-076	Frontend	4	UX-W2	UI/UX	1	3 sprints earlier

Task	Team	Sprint	Needs	From	Sprint	Gap
P3-I3	Backend	4	CT-14	AI	1	3 sprints earlier
PL-FRR4	Leads	4	BE-PK3	Backend	3	1 sprint earlier
PL-REL4	Leads	4	QA-RT4	QA	4	Same sprint
AI-058	AI	5	DEC-01	Backend	1	4 sprints earlier
AI-058	AI	5	DEC-02	Backend	1	4 sprints earlier
FE-033	Frontend	5	CT-05	Backend	1	4 sprints earlier
FE-033	Frontend	5	P3-I4	Backend	4	1 sprint earlier
FE-052	Frontend	5	P3-H1	Backend	4	1 sprint earlier
FE-052	Frontend	5	P3-H4	Backend	4	1 sprint earlier
FE-052	Frontend	5	UX-W3	UI/UX	1	4 sprints earlier
FE-053	Frontend	5	P3-H7	Backend	4	1 sprint earlier
FE-054	Frontend	5	P3-G4	Backend	4	1 sprint earlier
FE-054	Frontend	5	UX-W3	UI/UX	1	4 sprints earlier
FE-055	Frontend	5	P3-G1	Backend	3	2 sprints earlier
FE-055	Frontend	5	UX-W3	UI/UX	1	4 sprints earlier
FE-056	Frontend	5	P3-G3	Backend	4	1 sprint earlier
FE-056	Frontend	5	P3-I7	Backend	4	1 sprint earlier
FE-068	Frontend	5	P2-F3	Backend	3	2 sprints earlier
FE-068	Frontend	5	UX-W3	UI/UX	1	4 sprints earlier
FE-069	Frontend	5	P2-F3	Backend	3	2 sprints earlier
FE-071	Frontend	5	P2-F4	Backend	3	2 sprints earlier
FE-071	Frontend	5	UX-W3	UI/UX	1	4 sprints earlier
FE-072	Frontend	5	P2-F4	Backend	3	2 sprints earlier
FE-073	Frontend	5	P2-F6	Backend	3	2 sprints earlier
FE-078	Frontend	5	P2-F3	Backend	3	2 sprints earlier
MB-P56	Mobile	5	UX-W1	UI/UX	1	4 sprints earlier
P4-J1	Backend	5	DEC-07	AI	5	Same sprint
P4-K1	Backend	5	DEC-06	AI	1	4 sprints earlier
PL-FRR5	Leads	5	BE-PK4	Backend	4	1 sprint earlier
PL-REL5	Leads	5	QA-RT5	QA	5	Same sprint
AI-022	AI	6	QA-AI3	QA	4	2 sprints earlier
FE-057	Frontend	6	P4-J1	Backend	5	1 sprint earlier

Task	Team	Sprint	Needs	From	Sprint	Gap
FE-057	Frontend	6	P4-J3	Backend	5	1 sprint earlier
FE-057	Frontend	6	P4-J4	Backend	5	1 sprint earlier
FE-057	Frontend	6	UX-W4	UI/UX	1	5 sprints earlier
FE-057	Frontend	6	CT-20	Backend	1	5 sprints earlier
FE-059	Frontend	6	P4-K2	Backend	5	1 sprint earlier
FE-059	Frontend	6	UX-W4	UI/UX	1	5 sprints earlier
FE-059	Frontend	6	DEC-43	AI	6	Same sprint
FE-059	Frontend	6	CT-20	Backend	1	5 sprints earlier
FE-061	Frontend	6	P4-L1	Backend	5	1 sprint earlier
FE-062	Frontend	6	P4-L1	Backend	5	1 sprint earlier
FE-062	Frontend	6	UX-W4	UI/UX	1	5 sprints earlier
FE-101	Frontend	6	CT-02	Backend	1	5 sprints earlier
FE-101	Frontend	6	DEC-47	Backend	2	4 sprints earlier
MB-F4B	Mobile	6	P5-O1	Backend	5	1 sprint earlier
MB-F4B	Mobile	6	P5-O2	Backend	5	1 sprint earlier
MB-F4B	Mobile	6	DEC-47	Backend	2	4 sprints earlier
MB-F4B	Mobile	6	CT-07	Backend	1	5 sprints earlier
MB-F5	Mobile	6	P5-N2	Backend	5	1 sprint earlier
MB-F5	Mobile	6	CT-06	Backend	1	5 sprints earlier
MB-P54	Mobile	6	P5-O1	Backend	5	1 sprint earlier
MB-P54	Mobile	6	UX-W4	UI/UX	1	5 sprints earlier
P3-H6	Backend	6	DEC-03	AI	1	5 sprints earlier
P3-I8	Backend	6	CT-08	AI	1	5 sprints earlier
P4-K4	Backend	6	DEC-03	AI	1	5 sprints earlier
P4-K4	Backend	6	CT-13	AI	1	5 sprints earlier
P6-P3	Backend	6	CT-13	AI	1	5 sprints earlier
PL-FRR6	Leads	6	BE-PK5	Backend	5	1 sprint earlier
PL-REL6	Leads	6	QA-RT6	QA	6	Same sprint
FE-041	Frontend	7	P4-L4	Backend	6	1 sprint earlier
FE-041	Frontend	7	UX-W5	UI/UX	1	6 sprints earlier
FE-042	Frontend	7	P5-N1	Backend	6	1 sprint earlier
FE-043	Frontend	7	P4-J6	Backend	6	1 sprint earlier

Task	Team	Sprint	Needs	From	Sprint	Gap
FE-043	Frontend	7	UX-W5	UI/UX	1	6 sprints earlier
FE-058	Frontend	7	P4-J5	Backend	6	1 sprint earlier
FE-060	Frontend	7	P4-K3	Backend	5	2 sprints earlier
FE-060	Frontend	7	P4-K4	Backend	6	1 sprint earlier
FE-063	Frontend	7	P4-L2	Backend	6	1 sprint earlier
FE-063	Frontend	7	P4-L3	Backend	6	1 sprint earlier
FE-064	Frontend	7	P5-O7	Backend	6	1 sprint earlier
FE-065	Frontend	7	P5-O6	Backend	6	1 sprint earlier
FE-065	Frontend	7	UX-W5	UI/UX	1	6 sprints earlier
FE-067	Frontend	7	P5-M2	Backend	6	1 sprint earlier
FE-067	Frontend	7	UX-W5	UI/UX	1	6 sprints earlier
FE-070	Frontend	7	P5-M2	Backend	6	1 sprint earlier
FE-070	Frontend	7	UX-W5	UI/UX	1	6 sprints earlier
FE-074	Frontend	7	P6-P2	Backend	6	1 sprint earlier
FE-074	Frontend	7	UX-W6	UI/UX	1	6 sprints earlier
FE-074	Frontend	7	CT-20	Backend	1	6 sprints earlier
FE-075	Frontend	7	P6-P3	Backend	6	1 sprint earlier
FE-075	Frontend	7	P6-P4	Backend	6	1 sprint earlier
MB-P19	Mobile	7	P5-N1	Backend	6	1 sprint earlier
MB-P19	Mobile	7	UX-W4	UI/UX	1	6 sprints earlier
MB-P55	Mobile	7	P5-O4	Backend	5	2 sprints earlier
MB-P55	Mobile	7	P5-O3	Backend	6	1 sprint earlier
MB-P55	Mobile	7	UX-W4	UI/UX	1	6 sprints earlier
MB-P55	Mobile	7	DEC-39	Backend	6	1 sprint earlier
MB-P55	Mobile	7	DEC-40	Backend	6	1 sprint earlier
P6-R1	Backend	7	AI-080	AI	6	1 sprint earlier
PL-FRR7	Leads	7	BE-PK6	Backend	6	1 sprint earlier
PL-REL7	Leads	7	QA-RT7	QA	7	Same sprint
FE-077	Frontend	8	P6-R2	Backend	7	1 sprint earlier
FE-077	Frontend	8	UX-W6	UI/UX	1	7 sprints earlier
FE-077	Frontend	8	CT-20	Backend	1	7 sprints earlier
FE-109	Frontend	8	CT-05	Backend	1	7 sprints earlier

Task	Team	Sprint	Needs	From	Sprint	Gap
PL-FRR8	Leads	8	BE-PK7	Backend	7	1 sprint earlier
PL-REL8	Leads	8	QA-RT8	QA	8	Same sprint

SECTION 4 · APPENDIX

Future work

Kept and not built in the MVP: licence and SaaS by decision, items without a requirement, or items that need real operational data.

Team	Items	Description	Requirement
Backend	**P1-B7**	Central SaaS database connection, migrations and models	FR-SAAS
Backend	**P2-D5**	Licence guard (READ_ONLY and SETUP allowlists, 423) and plan-limit service (422)	FR-LIC-02 to 04
Backend	**P2-D6**	GET /license/state for the banner	FR-LIC-08
Backend	**P6-Q1 to Q3**	Port licence mode decision, check-in client, licence status page data	FR-LIC
Backend	**P7-S1 to S6**	Central SaaS core: admin sign-in, plans, port registry, licences, check-in API, platform audit	FR-SAAS-01 to 13
Backend	**P7-T1 to T2**	Billing transactions, fleet monitor and history browsing	FR-SAAS-12, 14, 15
Backend	**Post-MVP**	FCM wake-up push, manual licence token, signing-key rotation, bilingual errors, login lockout	FW-06, FW-09, FW-11
Frontend	**FE-008**	SaaS API contract and session mechanism	FR-SAAS
Frontend	**FE-028, FE-029**	Licence store, banners, write-action gating and 423 handling	FR-LIC-02, 08
Frontend	**FE-044**	P-22 License Status page	FR-LIC-09
Frontend	**FE-085 to FE-097**	Central SaaS Admin Dashboard (P-01 to P-15)	FR-SAAS
Frontend	**FE-104, FE-106, FE-116**	Read-only/setup tests, SaaS E2E, SaaS build	FR-LIC, FR-SAAS
Frontend	**FE-079**	P-61 Boat Management (blocked: no boat entity in FRD v3 or SQL)	FE-OQ-09
Frontend	**FE-080**	P-60 NCT Storage and Equipment	FR-MD-17, FW-05
Mobile	**Licence items**	READ_ONLY banner, token-expiry banner, licence state on P-53/P-54/P-55, READ_ONLY test	FR-LIC-08
Mobile	**NCT supervisor**	NCT supervisor role content on P-54/P-55	FW-05
AI	**AI-046**	A6 forecast uncertainty margin (zero-margin default shipped)	FR-ENV-08, FW-04
AI	**AI-100 to AI-103**	Training datasets and LightGBM M2, M3, M4 models	FR-RC-11, FW-03
AI	**AI-104**	C3 guarded automatic retraining	FR-LM-06, FW-01
AI	**AI-105**	C4 forecast bias correction	FR-LM-07, FW-02
AI	**AI-106**	Drift baselines and rollback checks for new models	FW-01
UI/UX	**P-01 to P-15, P-22**	SaaS Admin screens and licence status screen (kept in the Screens Specification future-work section)	FR-SAAS, FR-LIC
Platform	**FW-07, FW-08**	MDM, VPN, firewall, split DNS automation; TOS, PMIS, Nafeza and request adapters	FW-07, FW-08